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Association Between Otome Game Playtime and Defecation Behavior and Fecal Chemical Indicators

乙女向游戏游玩时长与排便行为及粪便化学指标的关联

Sjw、Syw

Sjw、Syw、zxc

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Abstract

Motivation: 构建并演示一套可复现的研究框架，用于评估乙女向游戏（乙游）游玩时长与排便行为（排便频次、单次排便时长、粪便形态）及粪便化学指标（含水率、pH、短链脂肪酸、支链脂肪酸、氨）之间的统计关联。 基于合成样本（n=1,480，18–39 岁），模拟 14 天行为日记与问卷；其中粪便化学子样本 n=633。暴露变量为乙游周游玩时长（小时/周），按四分位分组并作为连续变量建模。结局变量包括每周排便次数、单次排便时长、Bristol Stool Form Scale（BSFS）分值、硬便比例（BSFS 1–2）及复合“便秘样”指标，并在化学子样本中模拟含水率、pH、乙酸/丙酸/丁酸、支链脂肪酸与氨。采用多变量线性回归与 Logistic 回归，控制年龄、BMI、膳食纤维、饮水、睡眠、步数、压力及“如厕刷手机评分”，并进行分层、敏感性与（示例性）中介分析。 To demonstrate a reproducible analytic framework evaluating whether weekly otome-game playtime is statistically associated with defecation behavior (frequency, toilet duration per episode, stool form) and fecal chemical indicators (water content, pH, short-chain fatty acids, branched-chain fatty acids, ammonia). A synthetic cohort (n=1,480; ages 18–39) was generated to emulate a 14-day diary-plus-questionnaire study, including a stool-chemistry subsample (n=633). Exposure was weekly otome playtime (hours/week), analyzed by quartiles and continuously. Outcomes included bowel movements per week, toilet duration, Bristol Stool Form Scale (BSFS), hard-stool prevalence (BSFS 1–2), a composite constipation-like indicator, and chemistry markers. Multivariable linear and logistic regressions were fitted with adjustment for age, BMI, dietary fiber, water intake, sleep, step counts, perceived stress, and a phone-use-on-toilet score; subgroup, sensitivity, and illustrative mediation analyses were conducted.

补充说明：从行为健康研究角度，数字娱乐的“沉浸时长”并不只是屏幕时间本身，而常常与久坐结构、碎片化进食、饮水不足、睡眠后移及情绪波动等共同出现。乙游作为长线运营产品，其“活动周—日常周”的节律与“抽卡—保底”机制会造成时长的峰谷变化，适合用作生活方式暴露的一个可操作代理变量。本模板强调暴露测量的可复现性：真实研究建议同时记录系统屏幕使用时长、游戏内统计（若可得）及自报问卷，并在日记中区分‘排便用力’与‘如厕坐位停留’，以避免将刷手机导致的延长误判为生理性排便困难。

Results: 高时长组（Q4，均值 11.96 h/周）较低时长组（Q1，均值 2.37 h/周）单次排便时长更长（9.19 vs 6.75 min）、BSFS 更低（3.33 vs 3.91）、硬便比例更高（21.1% vs 8.6%）、复合便秘样比例更高（25.4% vs 8.6%）。在调整后模型中，乙游时长每增加 10 h/周与单次排便时长增加 1.618 min 相关；与每周排便次数减少 0.389 次相关；与复合便秘样风险升高相关（OR 3.768）。化学子样本显示含水率下降、pH 升高、丁酸下降与氨升高的方向一致模式。 Compared with Q1 (mean 2.37 h/week), Q4 (mean 11.96 h/week) showed longer toilet duration (9.19 vs 6.75 min), lower BSFS (3.33 vs 3.91), higher hard-stool prevalence (21.1% vs 8.6%), and higher composite constipation-like prevalence (25.4% vs 8.6%). In adjusted models, each 10 h/week increase in playtime was associated

with +1.618 minutes in toilet duration, -0.389 bowel movements per week, and increased odds of the composite outcome (OR 3.768). The chemistry subsample displayed coherent directional patterns: lower fecal water, higher pH, lower butyrate, and higher ammonia.
进一步地，结果提示：即便在模型中控制“如厕刷手机评分”等关键行为路径后，乙游时长与排便表现的关联仍存在一定残余效应，说明除厕所内停留之外，久坐、作息与饮食结构可能共同参与了表型形成。化学指标方面，含水率下降与 pH 上移可与更干硬的便形相互印证；丁酸与氨的方向性变化则为‘发酵底物改变—代谢谱轻度漂移’提供了讨论空间。需要强调：上述结果仅用于展示写作结构，不能替代真实人群证据。
Availability: This is a simulation study based on generated data; no human-subject data are included.
Contact: (email not provided)
Supplementary information: None.

1. Introduction and Background

1.1 背景与研究必要性（Background and Rationale）

近年来，移动端数字娱乐呈现出“高沉浸—高频触达—高粘性循环”的结构性特征。乙女向游戏（乙游）以强叙事、角色陪伴、周期性活动、概率性抽取（俗称“抽卡”）等机制，形成了稳定且持续的注意力投入。与传统“单机—通关”模式不同，乙游的时间结构更接近长期运营产品：日常任务、限时活动、联动上新与剧情更新构成了持续的外部提示，促使玩家在工作/学习之外形成碎片化但累积显著的使用时长。在中文互联网语境中，乙游玩家群体衍生出一套高度凝练的表达体系，例如“上头”（难以自控地沉浸）、“我推”（我支持/偏爱的角色）、“保底”（概率机制中的最低保障阈值）、“爆肝”（活动周高强度投入）、“回血”（情绪恢复/被治愈）、“吃谷”（购买周边，“谷”指 goods 的音译）、“梦女”（将自我代入角色关系的幻想叙事）等。这些“梗”既是文化符号，也能被视为行为线索：它们往往对应更高的情绪唤起、更高的时间集中度、更强的社交讨论与更明显的线下消费延伸。因此，在行为流行病学框架下，乙游并非单纯“玩游戏”，而是一组可拆解为‘时间投入—情绪调节—社交参与—消费外溢’的复合暴露。

此外，乙游的情绪功能（陪伴、治愈、代入）可能使其与压力事件高度耦合：压力越高，越可能通过游戏进行情绪调节；而情绪调节又可能伴随更晚的入睡、更少的运动与更不规律的饮食。换言之，乙游时长既可能是原因变量，也可能是压力与作息紊乱的表征变量。在论文写作中，提前承认这种双向性，并通过设计（分层、个体内比较）与统计控制（压力、睡眠、活动量）来约束解释边界，会让论证更接近行为流行病学的规范表达。

从时间预算的角度，乙游的占用并不完全是连续大块时间，更多发生在‘可被插入’的空隙：通勤、课间、睡前、以及如厕等。与传统“开局—通关”单机不同，乙游的日常任务与体力机制会推动玩家形成多次短时登录，叠加后形成较高周时长。研究上，这种碎片化结构意味着：仅用单一问卷回忆平均时长容易低估峰值，也容易忽略“活动周”带来的短期暴露冲击，因此更合适的做法是结合 14 天日记或连续屏幕时长记录来捕捉波动。

在乙游社区的日常交流中，许多高频词汇实际上指向可测量的行为成分。例如，“爆肝”往往对应活动周内连续多日的高时长在线；“保底”“歪了”“吃井”对应在抽卡不确定性下反复登录与持续投入；“氪金/课金”“吃谷”则反映出线上娱乐向线下消费的外溢；“欧皇/非酋”“上供”“发癫”更多关联情绪唤起与挫折-补偿循环。把这些语汇视为文化噪声会错过其行为指示意义，而把它们转化为结构化条目（频次、强度、场景）则能显著提高暴露刻画的分辨率。

在跨文化语境中，乙游玩家也会借用日语与英语用语（如“推し/oshi”“单推”“多推”“入坑/退坑”“课表式肝度”），这些词汇同样可以被视为‘投入结构’的标签：它们暗示玩家的注意力对象是否集中、是否存在长期追更与持续社群互动。若未来研究能把语汇使用与客观行为日志（登录次数、活动参与度、消费记录）进行关联验证，将有助于建立更稳健的构念测量链条。

Structural characteristics. Otome-oriented games (hereafter “otome games”) are characterized by strong narrative design, character-based companionship, cyclical events, and probabilistic reward mechanisms (commonly referred to as “gacha”). Together, these features generate sustained and stable attentional engagement. Unlike the traditional “single-player–completion” model, the temporal structure of otome games more closely resembles that of long-term live-service products: daily quests, limited-time events, cross-brand collaborations, and episodic narrative updates function as recurring external cues that prompt players to engage outside of work or academic obligations. This engagement often occurs in fragmented intervals yet accumulates into substantial overall usage time. Within the Chinese internet context, the otome player community has developed a highly condensed lexicon, including terms such as “shangtou” (intense, difficult-to-control immersion), “wo tui” (one’s favored character), “baodi” (a guaranteed minimum threshold in probabilistic systems), “baogan” (high-intensity participation during event periods), “huixue” (emotional recovery or healing), “chigu” (purchasing merchandise; “gu” as a phonetic rendering of “goods”), and “mengnü” (self-insert romantic fantasy involving a character). These expressions function not merely as cultural markers but also as behavioral indicators: they often correspond to heightened emotional arousal, increased temporal concentration, intensified social discussion, and spillover into offline consumption. Accordingly, within a behavioral epidemiology framework, participation in otome games should not be reduced to “playing a game,” but rather conceptualized as a composite exposure that can be analytically decomposed into time investment, emotional regulation, social participation, and consumption spillover.

Moreover, the affective functions of otome games—companionship, emotional comfort, and narrative identification—may render them highly coupled with stress-related events. Elevated stress levels may increase the likelihood of using the game for emotional regulation; such regulation, in turn, may be associated with later bedtimes, reduced physical activity, and more irregular dietary patterns. In this sense, gameplay duration may function both as a causal variable and as an indicator variable reflecting stress and disrupted daily routines. In formal academic writing, explicitly acknowledging this bidirectionality in advance, and constraining interpretive boundaries through design strategies (e.g., stratification, within-individual comparisons) and statistical controls (e.g., stress level, sleep duration, physical activity), would align the argument more closely with established standards in behavioral epidemiology.

From a time-budget perspective, engagement with otome games does not necessarily consist of prolonged, uninterrupted sessions; rather, it frequently occurs within “insertable” intervals, such as commuting,

breaks between classes, pre-sleep periods, and other routine gaps. In contrast to traditional single-player formats structured around discrete start-to-finish sessions, the daily task systems and stamina mechanics typical of otome games incentivize multiple short logins throughout the day, which cumulatively produce high weekly exposure. Methodologically, this fragmented structure implies that reliance on single retrospective self-report measures of average duration may underestimate peak exposure and fail to capture short-term surges associated with event weeks. A more appropriate approach would integrate 14-day diary methods or continuous screen-time tracking to capture intra-individual fluctuations.

Within community discourse, many high-frequency terms correspond to measurable behavioral components. For example, "baogan" often denotes sustained high-duration online engagement across consecutive days during event periods; "baodi," "waile" (missing the intended outcome), and "chijing" (reaching a hard pity threshold) reflect repeated logins and continued investment under probabilistic uncertainty; "kejin" (monetary spending) and "chigu" indicate spillover from online engagement to offline consumption; while "ouhuang" (exceptional luck), "feiqiu" (persistent bad luck), "shanggong," and "fadian" are more closely associated with emotional arousal and frustration–compensation cycles. To dismiss such terminology as cultural noise would overlook its behavioral signaling function; conversely, transforming these expressions into structured indicators—operationalized by frequency, intensity, and situational context—can substantially improve the resolution of exposure measurement.

In cross-cultural contexts, otome players frequently appropriate Japanese and English terms (e.g., "oshi," "single stan," "multi stan," "entering the fandom," "leaving the fandom," "schedule-style grinding intensity"). These lexical practices may likewise be interpreted as labels of engagement structure, signaling whether attentional focus is concentrated on a specific character and whether long-term narrative following and sustained community interaction are present. Future research that links linguistic usage patterns with objective behavioral logs (e.g., login frequency, event participation rates, expenditure records) would contribute to the development of a more robust construct validation framework.

1.2 排便行为为何适合作为数字沉浸暴露的终点（Why bowel outcomes）

排便行为是生活方式变化的敏感指标之一，原因在于其同时受到饮食结构（纤维与水分）、运动水平、睡眠节律、心理压力以及如厕行为习惯等多因素影响。与某些需要长期累积才出现差异的慢性终点相比，排便频次、单次排便时长、粪便形态等指标对短期行为扰动更敏感，适合用于观察"久坐—作息—饮食—注意力占用"这一类综合暴露的即时反应。临床与公共健康资料通常将便秘描述为：每周排便少于 3 次、粪便干硬/块状、排便困难或排不尽感等。在群体研究中，研究者常用 Bristol Stool Form Scale（BSFS）将粪便形态分为 1–7 型，作为可操作的自评表型；同时辅以频次与排便困难程度，可以更接近真实的功能性变化。需要强调的是，BSFS 作为自评量表存在主观误差，因此适合与频次、时长、饮水与膳食纤维等变量联合建模，而不宜被用作单一诊断。

从测量学角度，排便相关指标适合采用短周期重复记录：14 天日记能够在参与负担可控的前提下，捕捉到周末/工作日差异、活动周峰值与短期饮食波动对排便的即时反应。相比一次性回忆问卷，日记数据更适合构建时间序列特征（如连续未排便天数、便意延迟、排便用力的日内波动），也更适合后续采用混合效应模型进行个体内推断。

与频次和便形相比，"单次如厕时长"更像一个'生理-行为混合指标'：它既可能上升于排便困难，也可能上升于厕所内分心活动（刷手机、看视频、读文章）。近期关于"如厕用手机"的研究更多从痔疮风险讨论其危害，提示延长坐厕本身可能带来不良后果；因此，本研究框架把如厕手机使用作为关键路径变量单独测量，目的就是避免把行为停留误读为排便困难。

便秘并非单一疾病名称，而是由一组症状描述的功能性问题。临床与公共健康资料通常强调：排便频次减少（例如每周少于 3 次）、粪便干硬、排出困难或排不尽感等维度需要综合判断。罗马（Rome）标准对功能性便秘给出更细的条目化判据（如用力、硬便比例、阻塞感等），而 BSFS 则提供了在群体研究中更易实施的便形表型工具；既有研究也提示 BSFS 在健康人群中具备可接受的信度与效度，但在相邻类型（如 2/3、5/6）上仍可能存在主观误差。

此外，便秘相关结局的定义应尽量避免"单指标裁决"。例如，一些人排便频次不低，但持续存在用力与排不尽感；另一些人频次偏低却自觉正常。将频次、便形、用力程度与排不尽感组合成复合指标，既能提高表型的可解释性，也能在统计上减少因单一阈值导致的误分类。对于如厕时长，建议在问卷中增加"通常你会在马桶上做什么"的选项，以区分阅读/刷视频/打游戏等不同注意力占用行为。

Defecation behavior represents a sensitive indicator of lifestyle variation, as it is simultaneously influenced by multiple determinants, including dietary composition (particularly fiber and fluid intake), physical activity level, sleep–wake rhythm, psychological stress, and toileting habits. Compared with chronic endpoints that require prolonged accumulation before observable divergence emerges, indices such as bowel movement frequency, duration of a single defecation episode, and stool form respond more rapidly to short-term behavioral perturbations. These characteristics render them suitable for examining the immediate effects of composite exposures such as sedentary behavior, circadian disruption, dietary irregularity, and attentional displacement. Clinical and public health sources commonly define constipation as fewer than three bowel movements per week, the presence of hard or lumpy stools, difficulty during evacuation, or a sensation of incomplete emptying. In population-based research, investigators frequently employ the Bristol Stool Form Scale (BSFS), which classifies stool form into seven types, as an operationalizable self-reported phenotype. When combined with frequency and perceived difficulty, this measure more closely approximates functional changes. It should be noted, however, that as a self-report instrument, the BSFS is subject to subjective misclassification; it is therefore more appropriate for joint modeling with frequency, duration, hydration, and dietary fiber variables than for use as a standalone diagnostic criterion.

From a measurement perspective, bowel-related indicators are well suited to short-cycle repeated recording. A 14-day diary design can capture weekday–weekend differences, event-related exposure peaks, and short-term dietary fluctuations while maintaining manageable participant burden. Compared with a single retrospective questionnaire, diary data are better positioned to construct time-series features, such as consecutive days without defecation, latency of defecatory urge, and intra-day variability in straining. They also provide a stronger foundation for within-individual inference using mixed-effects models.

Relative to frequency and stool form, the duration of a single toileting episode constitutes a hybrid physiological–behavioral indicator. Prolonged duration may reflect evacuation difficulty, but it may also reflect attentional displacement within the restroom environment, such as smartphone use, video viewing, or reading. Recent studies addressing "smartphone use during toileting" have largely examined potential associations with hemorrhoidal risk, suggesting that prolonged sitting itself may entail adverse consequences. Accordingly, in the present

analytical framework, smartphone use during toileting is treated as a distinct pathway variable to prevent misclassification of behavioral delay as physiological dysfunction.

Constipation should not be conceptualized as a singular disease entity but rather as a functional condition defined by a constellation of symptoms. Clinical and public health guidelines emphasize the need for integrative assessment across dimensions, including reduced frequency (e.g., fewer than three bowel movements per week), hard stool consistency, excessive straining, and incomplete evacuation. The Rome criteria provide more granular diagnostic criteria for functional constipation, including straining, proportion of hard stools, and sensations of obstruction. In contrast, the BSFS offers a more feasible phenotypic tool for large-scale population studies. Existing literature suggests that the BSFS demonstrates acceptable reliability and validity in general populations, although subjective ambiguity may persist between adjacent categories (e.g., Types 2 versus 3, or Types 5 versus 6).

Finally, outcome definitions for constipation-related endpoints should avoid reliance on single-indicator thresholds. For example, some individuals may report normal frequency yet experience persistent straining and incomplete evacuation, whereas others may have lower frequency but no subjective discomfort. Constructing composite indicators that integrate frequency, stool form, straining severity, and incomplete evacuation can enhance phenotypic interpretability and reduce misclassification arising from rigid cutoffs. With respect to toileting duration, questionnaires should include items specifying typical activities performed while seated (e.g., reading, video consumption, gaming) to differentiate physiological evacuation time from attentional engagement behaviors.

1.3 理论框架：乙游时长→行为路径→排便表现→化学指标（Conceptual model）

本研究框架将乙游游玩时长视为上游暴露，并假设其主要通过三条路径影响排便：（1）久坐与身体活动减少：长时间坐姿可能降低肠蠕动的外部促进因素，并与总体活动不足共同出现；（2）睡眠与节律扰动：活动周“爆肝”与夜间补剧情可能压缩睡眠，进而影响肠道昼夜节律与排便反射；（3）如厕行为改变：玩家在如厕时继续刷手机（看攻略、刷社区、过剧情）可能延长坐厕时间，增加“在马桶上停留”的行为成分，从而让‘排便时长’同时包含生理排出与行为停留两部分。在上述路径基础上，若行为模式持续存在，还可能在粪便化学指标上呈现轻度但方向一致的变化，例如含水率降低、pH 升高、丁酸下降、氨升高等。这些指标并不用于诊断疾病，而用于提供“与既有肠道代谢知识相容”的机制线索。

因此，本框架不将化学指标视为诊断工具，而将其定位为“方向一致性”证据：若观察到便形偏硬、含水率下降，同时伴随 pH 上移、丁酸偏低/氨偏高的趋势，那么在论证链条中可以更自然地讨论‘饮水/纤维不足—发酵底物变化—代谢谱漂移—排便表型改变’这一组可能路径。

粪便化学指标提供了一个与行为表型相互印证的层面。短链脂肪酸（SCFAs）尤其是丁酸，被广泛讨论为结肠上皮能量来源并与屏障功能、炎症调控相关；当可发酵碳水/纤维不足或饮食结构偏移时，SCFAs 谱可能发生改变。另一方面，在糖类底物不足时，肠道微生物的蛋白发酵可能增强，产生更多含氮代谢物（如氨）并影响粪便酸碱环境；粪便 pH 的轻度上移也常被用作发酵模式变化的间接线索。

在机制层面，久坐与活动不足被认为会降低肠道蠕动的外部刺激，而睡眠不足可能扰动肠道-大脑轴与昼夜节律，从而改变排便反射与主观便意。人群研究中也常观察到‘久坐时间长 + 睡眠不足’的组合与便秘风险更高的现象，这提示暴露可能并非线性叠加，而存在交互或阈值效应（例如在活动周高时长且睡眠显著缩短时，风险陡增）。

需要补充的是，化学指标与排便表型之间并非一一对应关系：同样的硬便表型可能来自不同机制（饮水不足、纤维不足、肠道动力不足或如厕延迟），而化学指标也可能因个体基线菌群差异而呈现较大离散。因此，真实研究若希望更接近机制解释，应考虑在化学子样本中同步收集饮食底物信息（例如抗性淀粉、可溶性纤维）以及至少一次的菌群结构测量（16S 或宏基因组），以便区分“底物变化”与“菌群差异”两类来源。

The present analytical framework conceptualizes duration of otome game engagement as an upstream exposure and posits that its influence on bowel function operates primarily through three pathways. First, prolonged sitting and reduced physical activity: extended sedentary posture may attenuate external stimuli that facilitate intestinal motility and frequently co-occurs with overall physical inactivity. Second, sleep and circadian disruption: intensive engagement during event periods (e.g., high-intensity participation weeks) and nighttime narrative consumption may compress sleep duration, thereby disturbing gastrointestinal circadian rhythms and defecatory reflexes. Third, alteration of toileting behavior: continued smartphone use during toileting (e.g., reading strategy guides, browsing community discussions, or progressing through narrative content) may prolong seated time, increasing the behavioral component of “time spent on the toilet,” such that measured defecation duration comprises both physiological evacuation and behavioral delay.

On the basis of these pathways, if such behavioral patterns persist, mild but directionally consistent changes may also be detectable in fecal chemical indices, including reduced water content, elevated pH, decreased butyrate concentration, and increased ammonia levels. These indicators are not intended for disease diagnosis; rather, they serve as mechanistic correlates that are compatible with established knowledge of intestinal metabolism. Accordingly, chemical indices are positioned not as diagnostic tools but as evidence of directional coherence. For example, if harder stool form and reduced water content are observed concurrently with upward shifts in fecal pH and trends toward lower butyrate and higher ammonia, the inferential chain linking insufficient hydration or fiber intake to altered fermentable substrates, metabolic profile shifts, and subsequent phenotypic changes in bowel patterns can be articulated more plausibly.

Fecal chemical measures provide a complementary layer of validation alongside behavioral phenotypes. Short-chain fatty acids (SCFAs), particularly butyrate, are widely recognized as principal energy substrates for colonic epithelial cells and are implicated in barrier integrity and inflammatory regulation. When fermentable carbohydrate or fiber intake is reduced, the SCFA profile may shift accordingly. Conversely, in conditions of limited carbohydrate substrate availability, microbial protein fermentation may increase, generating greater quantities of nitrogen-containing metabolites (e.g., ammonia) and influencing the fecal acid–base environment. A modest elevation in fecal pH is often interpreted as an indirect indicator of altered fermentation patterns.

At the mechanistic level, sedentary behavior and insufficient physical activity are thought to diminish external stimuli promoting gastrointestinal motility, while sleep deprivation may disrupt the gut–brain axis and circadian regulation, thereby altering defecatory reflexes and subjective urge perception. Population-based studies frequently report that the combination of prolonged sedentary time and insufficient sleep is associated with elevated constipation risk. This observation suggests that exposures may not operate in a purely additive fashion; instead, interaction or threshold effects may exist, such that risk increases disproportionately under conditions of both high engagement duration and substantial sleep restriction.

It should be emphasized that fecal chemical indices and bowel phenotypes do not exhibit a one-to-one correspondence. Identical hard-stool

phenotypes may arise from heterogeneous mechanisms, including inadequate hydration, insufficient fiber intake, impaired colonic motility, or delayed toileting behavior. Likewise, chemical markers may display substantial inter-individual variability attributable to baseline differences in gut microbial composition. To approach mechanistic interpretation more closely, empirical studies incorporating chemical subsamples should concurrently collect detailed dietary substrate data (e.g., resistant starch, soluble fiber) and at least one measure of microbial community structure (e.g., 16S rRNA sequencing or metagenomic profiling), thereby enabling differentiation between substrate-driven shifts and microbiota-related variation.

1.4 相关文献与知识空缺（Related literature and gaps）

在正式讨论乙游时长与排便之间的可能联系之前，有必要把“便秘/排便异常”在公共健康语境中的常用定义说明清楚。权威健康教育资料通常将便秘描述为：排便频次减少（例如每周少于 3 次）、粪便干硬或块状、排出困难或疼痛、以及排不尽感等；同时也强调个体差异较大，只有在偏离个人通常模式并造成困扰时才构成问题。与“疾病名称”不同，便秘更像一组症状集合，因此在群体研究中，最稳妥的策略是把频次、便形与排便体验（用力、排不尽）作为多维指标联合表征，而不是用单一阈值裁决。

在研究工具方面，Bristol Stool Form Scale（BSFS）由于直观、可自评、可跨研究比较而被广泛采用。既有验证研究提示，BSFS 在健康成人及部分临床人群中具有较好的信度与效度，但在临床决策边界附近（例如 2/3 与 5/6）仍可能出现误差；这意味着研究者应通过图示培训、示例题与重复测量降低主观偏差，并在统计上避免把某一类型的微小变化过度解释为“疾病发生”。因此，本研究框架将 BSFS 作为便形表型的核心组成，同时配合排便时长与硬便比例等指标，以提高稳健性。

与乙游相关的“久坐—节律—排便”链条并非凭空假设。大量生活方式研究在更一般的暴露（总久坐时间、屏幕时间、睡眠不足、体力活动不足）层面都观察到与便秘或排便不适相关的模式，尤其是当久坐与短睡眠共同出现时，风险可能更明显。对乙游而言，关键问题不在于“游戏类型”本身，而在于其典型的使用结构：活动周冲刺、睡前补剧情、以及在固定情境（通勤/床上/如厕）反复打开应用所形成的节律扰动与身体静止。

此外，“如厕刷手机”并非边缘行为，而是会系统性地改变“排便时长”这一指标的含义：坐在马桶上的总时长既包含生理排出所需时间，也包含注意力占用导致的停留时间。关于“厕所内使用智能手机”的横断面研究主要从痔疮风险或久坐压力角度讨论其危害，其核心启示是：手机会显著延长坐厕停留，并把本来短暂的排便情境扩展为持续刷屏情境。因此，若研究不测量这一行为，很容易把‘刷屏久坐’误判为‘排便困难’，从而夸大暴露效应。

在现有文献中，“游戏/屏幕时间与排便”研究多以总屏幕时间或一般电子游戏为暴露，很少区分乙游这种“强叙事、长期运营、活动周波动明显”的产品形态；同时，也少有研究把亚文化参与（梗语使用、社群讨论、吃谷等）作为可测量的参与结构。由此产生两类空缺：其一，缺少能同时捕捉时间投入与参与结构的暴露刻画；其二，缺少把行为表型与粪便化学指标放在同一框架中解释的写作范式。本稿的目标正是以合成数据演示一种可复现的结构，供真实数据替换与后续验证。

Before formally examining the potential association between otome game engagement duration and bowel outcomes, it is necessary to clarify how “constipation” or “abnormal defecation” is defined within public health discourse. Authoritative health education materials typically describe constipation as reduced bowel movement frequency (e.g., fewer than three times per week), hard or lumpy stools, difficulty or pain during evacuation, and a sensation of incomplete emptying. At the same time, these sources emphasize substantial inter-individual variability; deviation from one’s habitual pattern accompanied by subjective distress is generally required for the condition to be considered clinically meaningful. Unlike a discrete disease entity, constipation is better conceptualized as a constellation of symptoms. Accordingly, in population-based research, the most methodologically prudent strategy is to characterize frequency, stool form, and evacuation experience (e.g., straining, incomplete evacuation) as multidimensional indicators rather than relying on a single cutoff threshold.

With respect to measurement tools, the Bristol Stool Form Scale (BSFS) is widely adopted because it is intuitive, self-administered, and comparable across studies. Validation research suggests that the BSFS demonstrates acceptable reliability and validity among healthy adults and selected clinical populations. However, near category boundaries (e.g., Types 2 versus 3, and Types 5 versus 6), misclassification remains possible. This implies that researchers should mitigate subjective bias through visual training materials, example items, and repeated measurements, and should avoid overinterpreting minor shifts between adjacent categories as indicative of disease onset. In the present framework, the BSFS is treated as a core phenotypic indicator of stool form, supplemented by measures such as defecation duration and proportion of hard stools to enhance robustness.

The proposed “sedentary behavior–circadian disruption–defecation” pathway associated with otome game engagement is not speculative. A substantial body of lifestyle research, examining broader exposures such as total sedentary time, screen time, sleep deprivation, and insufficient physical activity, has documented associations with constipation or defecatory discomfort. Notably, risk patterns appear more pronounced when prolonged sitting co-occurs with short sleep duration. In the context of otome games, the central issue is not the genre per se but its characteristic usage structure: intensive engagement during event weeks, narrative consumption before sleep, and repeated app access in fixed contexts (e.g., commuting, in bed, or during toileting), all of which may contribute to rhythm disturbance and bodily immobility.

Furthermore, smartphone use during toileting is not a marginal behavior but one that systematically alters the interpretive meaning of “defecation duration.” Total time seated on the toilet encompasses both physiological evacuation time and attentional engagement time attributable to device use. Cross-sectional studies addressing smartphone use in restroom settings have largely focused on hemorrhoidal risk or pressure-related consequences of prolonged sitting. A consistent implication of this literature is that smartphone use substantially extends toilet-seated duration, transforming a typically brief physiological event into an extended screen-based episode. In the absence of direct measurement of this behavior, studies risk misclassifying “screen-related prolonged sitting” as “defecatory difficulty,” thereby inflating the estimated exposure effect.

Within the existing literature, research on “gaming/screen time and bowel outcomes” has typically operationalized exposure as total screen time or general video game use, rarely distinguishing product forms such as otome games characterized by strong narrative immersion, long-term live-service operation, and pronounced event-week fluctuation. Moreover, few studies have incorporated subcultural participation structures—such as specialized vocabulary use, community engagement, or merchandise consumption—as measurable components of exposure. This generates two notable gaps: first, the absence of exposure characterizations capable of simultaneously capturing time investment and participation structure; second, the lack of integrative frameworks that interpret behavioral phenotypes alongside fecal chemical indices. The present

manuscript aims to demonstrate, using synthetic data, a reproducible structural model that may later be populated with empirical data for validation and refinement.

1.5 研究问题与假设（Research questions and hypotheses）

基于上述框架，本研究的核心问题可表述为：在控制饮食纤维、饮水、睡眠、步数与压力等关键混杂后，乙游周游玩时长是否与排便行为（频次、单次如厕时长、便形/硬便）呈稳定关联？我们提出三条预设假设：（H1）乙游时长越高，单次如厕时长越长且硬便比例越高；（H2）乙游时长与每周排便次数呈负向关联，且在睡眠不足或活动周强度更高时效应更强（效应修饰）；（H3）如厕刷手机评分在“乙游时长→如厕时长延长”的关联中承担部分解释作用，但不能完全消除关联（部分中介）。

化学子样本属于探索性问题：在行为表型出现“偏干、偏慢”的模式时，粪便含水率是否下降、pH 是否上移、丁酸是否偏低、氨是否偏高，从而形成方向一致的“机制线索”组合？需要强调的是，化学指标在本框架中不用于医疗诊断，而用于回答“行为表型的方向是否与已知的肠道代谢常识相容”。因此，写作上我们将其定位为‘一致性证据’，并在讨论中明确其依赖饮食底物与样本处理信息的前提。

On the basis of the foregoing framework, the central research question may be formulated as follows: after controlling for key confounders, including dietary fiber intake, hydration, sleep duration, step count, and psychological stress, is weekly duration of otome game engagement stably associated with bowel-related outcomes (frequency, single-episode toileting duration, and stool form/hard stool proportion)?

Three a priori hypotheses are proposed. First (H1), greater weekly engagement duration will be associated with longer single-episode toileting duration and a higher proportion of hard stools. Second (H2), engagement duration will be negatively associated with weekly bowel movement frequency, and this association will be stronger under conditions of insufficient sleep or higher event-week intensity, indicating potential effect modification. Third (H3), a composite score reflecting smartphone use during toileting will partially account for the association between engagement duration and prolonged toileting time; however, the association will not be fully attenuated after adjustment, consistent with partial mediation rather than complete explanation.

The chemical subsample is positioned as exploratory. Specifically, when behavioral phenotypes exhibit a “drier, slower” pattern, do fecal water content decrease, pH shift upward, butyrate concentrations trend lower, and ammonia concentrations trend higher, thereby forming a directionally coherent set of mechanistic signals? It must be emphasized that, within this framework, chemical indicators are not intended for clinical diagnosis. Rather, they address a narrower inferential question: whether observed behavioral phenotypes are directionally compatible with established knowledge of intestinal metabolic processes. Accordingly, these indicators are framed as evidence of consistency. Their interpretation is explicitly conditioned on underlying dietary substrate information and sample handling parameters, both of which may substantially influence measured values.

1.6 研究设计与时间窗（Study design and time window）

本稿采用数据收集的方式，收集乙游玩家“14 天前瞻性日记 + 基线问卷 + 粪便化学子样本”的研究。选择 14 天的理由是：一方面可覆盖至少一个周末，捕捉作息差异；另一方面能在不显著增加参与负担的情况下获得足够的日内变异信息。日记通过手机表单或小程序完成，并辅以屏幕使用时长截图进行一致性校验。此外，为避免把‘上厕所刷手机’与‘排便困难’混为一谈，日记分别记录“排便是否发生”“排便用力程度”“如厕时手机使用时长”，从而在模型中分离行为停留与生理困难。

此外，在研究前预注册（pre-registration）数据清洗与异常值规则，例如：如厕时长超过某阈值（如 60 分钟）标记为极端值、排便为 0 却填写便形标记为不一致、饮水量与纤维摄入落在生理不合理范围时提示核对。预先定义规则并在附录报告剔除比例，可以显著提升论文的透明度。

为增强可复现性，在日记系统中加入自动时间戳与提醒机制（例如每天固定时段推送填报），并设定最小字段校验（如排便发生时才允许填写 BSFS 与时长）。对乙游时长的记录时区分“主动游玩”（剧情/操作）与“被动在线”（挂机/自动战斗），并记录是否处于活动周，以便在统计上把短期峰值暴露与基线暴露分开建模。

为降低遗漏与回忆偏差，日记记录可能影响排便的短期事件：是否旅行或跨时区、是否处于经期、是否出现感冒发热、是否服用铁剂/钙剂/止泻药/泻药等常见药物或补充剂，以及当天是否摄入大量咖啡因或酒精。这些因素在短窗口内就可能显著改变排便频次与便形，若不记录，容易把短期波动错误归因于乙游时长。

The present study adopts a structured data collection design consisting of a 14-day prospective diary, a baseline questionnaire, and a fecal chemical subsample among otome game players. The selection of a 14-day observation window is methodologically justified for two reasons. First, it ensures coverage of at least one weekend cycle, thereby capturing potential weekday–weekend differences in sleep and activity patterns. Second, it provides sufficient intra-individual variability while maintaining an acceptable participant burden. Daily diary entries are completed via a mobile-based form or mini-program, supplemented by screenshots of device-level screen time to enable consistency checks between self-reported and objectively recorded exposure.

To avoid conflating “smartphone use during toileting” with physiological defecatory difficulty, the diary instrument records the following variables separately: (1) whether defecation occurred, (2) perceived straining severity, and (3) duration of smartphone use while seated on the toilet. This structure allows the analytical model to disentangle behavioral prolongation from physiological evacuation difficulty.

Prior to data collection, pre-registration procedures define data cleaning protocols and outlier handling rules. For example, toileting duration exceeding a pre-specified threshold (e.g., >60 minutes) is flagged as an extreme value; cases reporting no bowel movement yet providing a stool form classification are marked as internally inconsistent; daily hydration or fiber intake values falling outside physiologically plausible ranges trigger verification prompts. Predefining such criteria and reporting exclusion proportions in an appendix materially enhances transparency and reproducibility.

To strengthen reliability, the diary system incorporates automated time stamps and reminder notifications (e.g., fixed daily prompts), along with minimal field validation rules (e.g., allowing Bristol Stool Form Scale entries and defecation duration fields only when defecation is reported). Otome game engagement duration is recorded with structural differentiation between “active play” (narrative progression, interactive gameplay) and “passive online presence” (e.g., auto-battle or idle states). Event-week status is also logged, permitting separate modeling of short-term peak exposure versus baseline engagement.

To reduce omission and recall bias, the diary additionally captures short-term events likely to influence bowel patterns. These include travel or time zone shifts, menstrual status, acute illness (e.g., fever or respiratory infection), and use of medications or supplements known to affect bowel function (e.g., iron, calcium, antidiarrheals, laxatives). Daily intake of high caffeine or alcohol quantities is also recorded. Each of these factors may substantially alter bowel frequency or stool form

within a short window; failure to account for them would risk misattributing transient fluctuations to variation in otome game exposure.

1.7 研究对象、纳入排除与伦理声明（Participants, eligibility, and ethics statement）

纳入对象为 18–39 岁成人，近 3 个月内至少游玩过 1 款乙游。排除标准包括：近 4 周使用抗生素（可能显著影响肠道代谢与化学指标）、明确的严重胃肠道疾病史、日记完成率不足 80%、或出现自相矛盾的记录（如排便次数为 0 但填写了 BSFS）。真实研究应在伦理审批后开展，重点包括：样本匿名化、可随时退出、粪便样本作为生物材料的安全处理与转运规范、以及明确声明研究不用于医疗诊断。

对于粪便化学子样本，除生物安全与运输规范外，还在方案中明确采样时间窗（例如晨便或任意一次排便）、采样到冷冻的最大允许时间、保存温度与冻融次数记录方式。由于这些前处理因素会影响读数，把其作为质控元数据进入分析，在附录中报告批次与处理流程。

伦理层面，涉及游戏行为与如厕/排便的调查具有一定隐私敏感性。强调数据最小化原则（只收集与研究问题直接相关的字段）、采用匿名化编号、并在知情同意中清晰说明数据用途、保存期限与销毁方式。若收集屏幕时长截图，避免包含通知内容或聊天信息，并提供遮挡示例以降低意外暴露风险。

另外，因为研究对象包含学生群体，在问卷中加入“课业/考试周负荷”与“熬夜补作业”条目，因为这些因素既可能提高乙游作为解压方式的使用，也可能通过饮食与睡眠改变排便状态。把学习压力显式量化，有助于在解释时避免把‘压力效应’误写成‘游戏效应’。

Eligible participants are adults aged 18–39 years who have played at least one otome game within the past three months. Exclusion criteria include antibiotic use within the preceding four weeks (given its potential to substantially alter gut metabolism and chemical indices), a documented history of severe gastrointestinal disease, diary completion rates below 80%, or internally inconsistent records (e.g., reporting zero bowel movements while providing a stool form classification). Any empirical implementation should be conducted following formal ethics approval. Core considerations include participant anonymization, the right to withdraw at any time, biosafety and transport protocols for fecal samples as biological material, and explicit clarification that the study is not intended for medical diagnosis.

For the fecal chemical subsample, the protocol further specifies pre-analytical conditions beyond general biosafety requirements. These include the sampling time window (e.g., first morning stool or any single defecation), the maximum allowable interval between collection and freezing, storage temperature, and documentation of freeze–thaw cycles. Because pre-analytical handling can materially influence measured values, these parameters are incorporated as quality-control metadata within the analytical dataset and reported transparently in the appendix, including batch information and processing workflows.

From an ethical perspective, surveys addressing gaming behavior and toileting/defecation involve heightened privacy sensitivity. The study adheres to a data minimization principle, collecting only variables directly relevant to the research question. Anonymous identifiers are employed, and the informed consent document clearly specifies data usage, retention duration, and destruction procedures. When collecting screen-time screenshots for exposure validation, participants are instructed to avoid capturing notifications or chat content, and masking examples are provided to reduce the risk of inadvertent disclosure.

Given that the target population may include students, the questionnaire additionally incorporates items assessing academic workload, examination periods, and late-night study patterns. These factors may increase reliance on otome games as a stress-regulation strategy while simultaneously altering sleep and dietary behaviors that influence bowel patterns. Explicit quantification of academic stress permits more accurate adjustment in statistical models and reduces the risk of attributing stress-related effects to gaming exposure.

Table 1. Emulated recruitment and quality-control flow.

Stage	n	Notes
Initial screening	1716	Synthetic pool
Excluded: age outside 18–39	39	Eligibility
Excluded: duplicate submissions	48	Device/account match
Excluded: diary completion <80%	86	Data quality
Excluded: antibiotic use (past 4 weeks)	42	Chemistry confound
Excluded: extreme outliers / inconsistent logs	21	Pre-defined rules
Included in main analysis	1480	Diary + questionnaire
Chemistry sub-sample	633	Usable synthetic chemistry panel

1.8 变量定义与测量细节（Measures and operationalization）

主要暴露变量为乙游周游玩时长（小时/周）。在真实研究中，可将其拆分为：剧情推进时长、日常任务时长、活动周冲刺时长与后台挂机/自动战斗时长，以降低不同游戏机制造成的不可比性。文化参与变量采用“梗语沉浸指数”（0–24），将“上头、我推、吃谷、保底、爆肝、回血、梦女、抽卡”八个条目按 0–3 评分累计。路径变量为“如厕刷手机评分”（0–10），由如厕时使用频率与平均使用时长合成。结局指标包括：每周排便次数、单次排便时长、BSFS 分型、硬便（BSFS 1–2）、以及复合便秘样指标（硬便或每周<3 次或单次≥12 分钟）。化学子样本的指标包括粪便含水率、pH、乙酸/丙酸/丁酸、支链脂肪酸与氨。协变量包括年龄、BMI、膳食纤维、水摄入、睡眠、步数与压力评分；这些变量之所以必须纳入，是因为它们同时影响游戏时长与排便/化学指标，若不控制将导致系统性混杂。

协变量测量建议尽量保持可执行性与覆盖关键混杂：饮食可使用短版 FFQ 聚焦纤维来源与零食结构，饮水用毫升或杯数换算，活动量可用手机健康数据或可穿戴步数，压力可用简短量表。通过‘低成本但关键覆盖’的策略，能在不显著增加负担的情况下减少系统性混杂。

变量操作化方面，本模板采用“梗语沉浸指数”与“如厕刷手机评分”作为对传统屏幕时长的结构化补充。真实研究可进一步验证其测量属性：例如通过重测信度（同一参与者在相邻周的稳定性）、

与客观行为指标的相关性（屏幕时长、社区浏览时长）以及与外部效标的关系（活动周参与、消费行为）来评估构念效度。

对于活动周暴露的编码，一种可行策略是把 14 天窗口内的‘活动冲刺日’计数（0–14 天）与主观强度（1–5）组合为“活动周强度指数”，并在模型中与总时长同时进入，以区分“同样总时长但集中度不同”的两类玩家。与此同时，可记录夜间游玩比例（如 22:00 后占比）作为节律扰动的代理变量，从而在统计上更直接地检验‘晚睡-排便变化’的路径假设。

The primary exposure variable is weekly duration of otome game engagement (hours per week). In empirical implementation, this measure may be disaggregated into narrative progression time, routine task time, event-week intensive play time, and background idle/auto-battle time. Such differentiation reduces non-comparability arising from heterogeneous game mechanics and allows more precise modeling of exposure structure.

Cultural participation is operationalized through a “meme immersion index” (range: 0–24), derived from eight items—such as expressions equivalent to immersive fixation, preferred character identification, merchandise purchasing, probabilistic guarantee thresholds, intensive event grinding, emotional recovery, self-insert fantasy identification, and gacha engagement—each scored on a 0–3 scale and summed. The pathway variable is a “toileting smartphone use score” (range: 0–10), constructed from reported frequency of device use during toileting and average duration per episode.

Outcome indicators include weekly bowel movement frequency, single-episode defecation duration, stool form as classified by the Bristol Stool Form Scale, hard stool phenotype (Types 1–2), and a composite constipation-like indicator defined as any of the following: hard stool, fewer than three bowel movements per week, or single-episode duration ≥12 minutes. The chemical subsample includes fecal water content, pH, acetate, propionate, butyrate, branched-chain fatty acids, and ammonia. Covariates include age, body mass index, dietary fiber intake, total water intake, sleep duration, step count, and perceived stress score. These variables are included because they plausibly influence both gaming duration and bowel or chemical outcomes; failure to adjust for them would introduce systematic confounding.

Measurement of covariates should balance feasibility with coverage of key confounders. Dietary intake may be assessed using a brief food frequency questionnaire focused on fiber sources and snack composition. Fluid intake can be recorded in milliliters or standardized cup equivalents. Physical activity may be derived from smartphone health applications or wearable step counts. Stress can be measured using a short validated scale. This “low-burden but critical coverage” strategy minimizes participant fatigue while reducing residual confounding.

From an operationalization perspective, the meme immersion index and toileting smartphone use score function as structured complements to conventional screen-time metrics. Future empirical studies may further evaluate their measurement properties through test–retest reliability (stability across adjacent weeks), correlation with objective behavioral indicators (e.g., device-recorded screen time, community browsing duration), and associations with external criteria (e.g., participation in event weeks, expenditure behaviors) to establish construct validity.

For coding event-week exposure, one feasible approach is to combine the number of “intensive event days” within the 14-day window (range: 0–14) with a subjective intensity rating (1–5) to generate an “event-week intensity index.” Entering this index alongside total engagement duration in statistical models permits differentiation between players with similar cumulative exposure but differing temporal concentration. Additionally, the proportion of gameplay occurring after 22:00 may be recorded as a proxy for circadian disruption, enabling more direct statistical testing of the hypothesized late-night engagement–bowel pattern pathway.

Table 2. Variable codebook.

Domain	Variable (CN)	Variable (EN)	Unit/Coding	Description
Exposure	乙游周游玩时长	Otome playtime	h/week	Primary exposure; continuous + quartiles
Culture	梗语沉浸指数	Meme Immersion Index	0–24	8 items scored 0–3 each
Pathway	如厕刷手机评分	Phone-on-toilet score	0–10	Composite frequency×duration
Lifestyle	膳食纤维	Dietary fiber	g/day	14-day mean
Lifestyle	饮水量	Water intake	mL/day	14-day mean
Lifestyle	睡眠时长	Sleep duration	h/day	14-day mean
Lifestyle	步数	Daily steps	steps/day	14-day mean
Lifestyle	压力评分	Perceived stress	1–10	Self-rated
Outcome	每周排便次数	BM/week	count	Diary-derived weekly count
Outcome	单次排便时长	Toilet duration	min/episode	Diary-derived mean
Outcome	粪便形态	BSFS	1–7	Diary-reported stool form
Outcome	硬便	Hard stool	0/1	BSFS 1–2
Outcome	复合便秘样	Composite constipation-like	0/1	Hard stool OR BM/week<3 OR duration≥12
Chemistry	含水率	Fecal water	%	Chemistry panel
Chemistry	pH	Fecal pH	-	Chemistry panel
Chemistry	丁酸	Butyrate	μmol/g	Chemistry panel
Chemistry	氨	Ammonia	mmol/kg	Chemistry panel

1.9 统计分析与稳健性策略（Statistical analysis and robustness）

统计分析分为四个层次：第一，描述性统计与趋势检验，比较四分位组的均值/比例并检验线性趋势；第二，多变量回归（连续结局用线性回归，二元结局用 Logistic 回归），以乙游时长（每+10 小时/周）为主解释变量，同时控制协变量；第三，分层分析，按睡眠不足与活动周强度分层检验效应异质性；第四，敏感性分析，包括剔除极端高时长、限制为非活动周、以及在化学子样本中重复估计。在写作层面，建议在真实研究里补充：缺失值处理策略（多重插补或完全案例分析）、共线性检查（VIF）、以及预先定义的异常值规则。

稳健性方面，建议至少报告：剔除极端高时长后的估计、限制为非活动周的估计、以及加入/移除关键路径变量（如如厕刷手机评分）后的系数变化。若存在非线性可能，可用限制性立方样条检验阈值，并在图中展示暴露-结局曲线以帮助读者直观理解。

统计建模上，乙游时长以每增加 10 小时/周作为单位，有助于解释效应大小并避免过小单位导致的系数难以解读。对计数结局（如 BM/week）在真实研究中可考虑负二项回归以处理过度离散；对 BSFS 可用有序 Logistic 或将硬便（1–2 型）作为二元结局。若采用逐日数据，更推荐混合效应模型以利用个体内信息并降低稳定个体差异带来的混杂。

在多终点研究中还需注意多重比较问题。真实研究若同时检验频次、时长、便形与多项化学指标，建议预先指定主要终点（primary outcome），其余作为探索性结果，并可采用控制错误发现率（FDR）或分层解释的方式，避免把偶然显著解释为稳定效应。对化学指标尤其如此：批次效应与前处理差异可能导致假阳性，因而应优先报告方向一致性与效应量大小，而不仅是 P 值。

示例性的中介分析（例如把如厕刷手机作为中介）在写作中应明确其前提：中介变量必须在时间上位于暴露之后、结局之前，并且不存在未测混杂同时影响中介与结局。若这些前提无法满足，建议把中介结果表述为“解释比例的统计分解”而非因果路径证据，并将其定位为生成性假设，为后续实验或干预研究提供方向。

The statistical analysis is organized into four tiers. First, descriptive statistics and trend tests are used to compare means and proportions across quartiles of exposure and to evaluate linear trends. Second, multivariable regression models are fitted, using weekly otome game duration as the primary explanatory variable (per +10 hours/week) while adjusting for prespecified covariates; linear regression is applied for continuous outcomes and logistic regression for binary outcomes. Third, stratified analyses are conducted to assess effect heterogeneity, with strata defined by insufficient sleep and event-week intensity. Fourth, sensitivity analyses are performed, including exclusion of extreme high-duration observations, restriction to non–event-week periods, and re-estimation within the chemical subsample. In reporting, empirical studies should additionally specify missing-data handling (e.g., multiple imputation versus complete-case analysis), collinearity diagnostics (e.g., variance inflation factors), and prespecified outlier rules.

For robustness, at minimum the following should be reported: estimates after excluding extreme high-duration observations; estimates restricted to non–event-week exposure; and coefficient changes when key pathway variables (e.g., toileting smartphone use score) are added to or removed from the model. Where nonlinearity is plausible, restricted cubic splines may be used to test for threshold effects, and exposure–outcome curves should be presented to facilitate interpretability.

With respect to model specification, scaling otome game duration per 10 hours/week improves interpretability and avoids coefficients that are difficult to contextualize when the unit is too small. For count outcomes (e.g., bowel movements per week), negative binomial regression may be preferable in the presence of overdispersion. For stool form, either an ordinal logistic model for the full BSFS distribution or a binary specification focusing on hard stool (Types 1–2) is appropriate. If day-level observations are analyzed, mixed-effects models are recommended to leverage within-individual variation and to reduce confounding by stable person-level characteristics.

Given the multi-endpoint nature of the study, multiple comparison considerations require explicit attention. If frequency, duration, stool form, and multiple chemical indices are tested concurrently, it is advisable to prespecify a primary outcome and treat the remaining outcomes as exploratory. Approaches such as false discovery rate control or hierarchical interpretation can reduce the risk of overinterpreting chance findings as stable effects. This is particularly important for chemical indices, where batch effects and pre-analytical variability can inflate false positives; consequently, emphasis should be placed on directionality and effect sizes rather than p-values alone.

Finally, any illustrative mediation analysis (e.g., toileting smartphone use as a mediator) should state its identifying assumptions. Specifically, the mediator must temporally follow the exposure and precede the outcome, and there must be no unmeasured confounding affecting both the mediator and the outcome. If these conditions cannot be credibly satisfied, mediation results should be described as a statistical decomposition of explanatory proportion rather than as evidence of a causal mechanism, and positioned as hypothesis-generating guidance for subsequent experimental or intervention studies.

1.10 样本量、代表性与统计功效（Sample size, representativeness, and power）

在研究中，样本量规划围绕“可检测的最小效应量”与“重复测量结构”展开。若终点为每周排便次数或单次如厕时长等连续变量，可基于预期标准差与目标差异（例如时长差异 1.0–1.5 分钟）进行功效估算；若终点为硬便或复合便秘样等二元变量，则应基于基线发生率与可接受的相对风险/OR 进行估算。由于 14 天日记提供了个体内多次观察，混合效应模型往往能提高统计效率，使得在相同样本量下更容易识别与时间相关的变化；但与此同时，也预留因日记不完整、极端值剔除与化学子样本缺失带来的有效样本损失。

代表性方面，乙游玩家并非随机抽样的一般人群：年龄分布、性别比例、学习/工作压力水平与社交媒体使用强度可能与整体人群不同。因此，研究目标应清晰界定为“乙游玩家内部的剂量-反应关联”或“在玩家群体中的风险画像”，而不宜在缺乏加权或多源招募时直接外推到全人群。实践中可采用分层招募策略（不同城市、不同学校/职业、不同游戏平台），并在分析中使用后分层权重或敏感性分析评估外推稳定性。

统计功效之外，还详细规划了“结果呈现的可解释性”。当复合结局的基线发生率较低时，OR 可能显得较大，但其绝对风险差可能很小；因此预先定义主要呈现指标：回归系数/OR 之外，同时报告调整后的预测概率、风险差与分位数差异（如中位数如厕时长差）。这种写法能降低读者对‘显著但不重要’的误解，也更符合公共健康研究的沟通习惯。

Sample size planning in this study is structured around the minimum detectable effect size (MDES) and the repeated-measures design. For continuous outcomes such as weekly bowel movement frequency or single-episode toileting duration, power calculations can be based on an anticipated standard deviation and a target mean difference (e.g., a 1.0–1.5 minute difference in duration). For binary outcomes such as hard stool or a composite constipation-like indicator, calculations should instead rely on the baseline event rate and a prespecified, clinically or practically meaningful relative effect (e.g., an odds ratio or relative risk deemed detectable and relevant). Because the 14-day diary yields multi-

ple within-person observations, mixed-effects models often improve statistical efficiency, increasing power to detect time-linked changes at a given sample size. However, the design should explicitly budget for reductions in effective sample size due to incomplete diary compliance, exclusion of extreme values, and missingness in the fecal chemistry subsample.

With respect to representativeness, otome game players do not constitute a random draw from the general population. Their age distribution, gender composition, academic or occupational stress levels, and social media intensity may systematically differ from population norms. Consequently, the study objective should be stated explicitly as estimating dose–response associations within the otome player population or characterizing risk profiles among players, rather than making unqualified inferences about the general population in the absence of weighting strategies or multi-source recruitment. In practice, stratified recruitment can be implemented across cities, school versus workforce settings, and gaming platforms, while post-stratification weights or sensitivity analyses can be used to evaluate the stability of inferences under alternative assumptions about generalizability.

Beyond statistical power, the protocol also prioritizes interpretability of result presentation. When baseline prevalence of a composite binary outcome is low, odds ratios may appear large even when the absolute risk difference is small. To reduce the likelihood of "statistically significant but practically trivial" interpretations, key reporting metrics should be prespecified. In addition to regression coefficients or odds ratios, results should include adjusted predicted probabilities, absolute risk differences, and distributional contrasts such as quantile differences (e.g., adjusted median differences in toileting duration). This reporting strategy is more consistent with public health communication norms and improves the substantive interpretability of estimated effects.

1.11 粪便样本采集与化学检测流程（Stool sampling and chemistry workflow）

化学子样本的核心挑战在于前处理与保存条件的可控性。若希望测量 SCFAs 与氨等易受降解影响的指标，理想流程是：参与者在规定时间窗内采集少量样本，使用一次性采样勺置入预标号的密封管；在家用冷冻或冷藏短时保存后，于 24 小时内进入冷链转运并在实验室-80°C 长期保存。采样到冷冻的间隔、运输温度、冻融次数与检测批次都应被记录为质控元数据，并在统计模型中作为协变量或分层因素处理。

在写作中，检测方法建议至少交代：检测平台（例如 GC-MS/LC-MS 或色谱法）、内标与校准曲线、定量单位换算（μmol/g 等）、以及重复测量或空白对照的安排。若样本量允许，应报告批内/批间变异系数（CV）与最小可检测限（LOD），并给出剔除规则（如超出 LOD、显著偏离批次分布）。这些细节能显著提升读者对"化学指标可解释性"的信任。

需要进一步强调的是，化学子样本往往存在"选择偏差"：愿意提供粪便样本的参与者可能在健康意识、生活方式或症状体验上与未提供者不同。因此，建议在主样本中比较"是否进入化学子样本"的基线特征，并在附录中报告差异；必要时可使用逆概率加权（IPW）或倾向评分匹配来降低选择偏差对化学结论的影响。

The principal methodological challenge of the fecal chemical subsample lies in the controllability of pre-analytical handling and storage conditions. For analytes such as short-chain fatty acids (SCFAs) and ammonia, which are susceptible to degradation and environmental influence, an optimized workflow is required. Participants should collect a small stool aliquot within a prespecified time window, transfer it using a disposable sampling spoon into a pre-labeled, airtight tube, and store it temporarily under home refrigeration or freezing conditions. The sample should then enter a cold-chain transport system within 24 hours and be stored at −80°C upon arrival at the laboratory. Key pre-analytical parameters—including the interval from collection to freezing, transport temperature, number of freeze–thaw cycles, and analytical batch—should be documented as quality-control metadata. These variables may be incorporated into statistical models as covariates or handled through stratified analyses to mitigate measurement bias.

In manuscript preparation, the analytical methodology should be described with sufficient specificity to ensure reproducibility. At minimum, this includes the detection platform (e.g., gas chromatography–mass spectrometry [GC–MS], liquid chromatography–mass spectrometry [LC–MS], or chromatographic quantification methods), use of internal standards and calibration curves, unit standardization (e.g., μmol/g wet weight), and procedures for replicate measurements or blank controls. Where sample size permits, reporting intra-batch and inter-batch coefficients of variation (CVs) and limits of detection (LODs) enhances interpretability. Predefined exclusion criteria—for example, values below LOD or extreme deviations from batch-specific distributions—should be transparently specified. Such methodological disclosure materially strengthens confidence in the validity and interpretability of chemical findings.

An additional concern is potential selection bias in the chemical subsample. Participants willing to provide fecal specimens may systematically differ from non-providers in health awareness, lifestyle behaviors, or symptom burden. Accordingly, baseline characteristics of participants included versus excluded from the chemical subsample should be compared and reported, preferably in an appendix. If meaningful differences are observed, analytical strategies such as inverse probability weighting (IPW) or propensity score–based adjustment may be employed to reduce selection-related distortion in chemical outcome estimates.

1.12 数据治理、隐私与可复现性（Data governance, privacy, reproducibility）

由于研究同时涉及数字行为（游戏时长、屏幕使用）与生物材料（粪便样本），数据治理应从一开始就设计为"最小必要原则"。在数据收集层面，只收集回答研究问题所必需的字段；对屏幕时长截图等潜在敏感信息，应提供遮挡指引并在上传端进行自动去标识化（例如移除通知栏、联系人名）。在存储层面，使用分离式映射表保存个人联系信息与研究数据，并设置访问权限与审计日志。

可复现性方面，建议将变量字典、清洗规则、以及最小可运行分析脚本作为补充材料公开；若不能公开原始数据，也应提供可复现实验的合成数据生成脚本与参数范围，并在论文中说明合成规则与随机种子策略。对于本文这种"先用合成数据演示框架"的写法，最关键的是让读者能在不接触隐私数据的情况下复现表格结构、模型公式与报告逻辑。

Because the study involves both digital behavioral data (e.g., gameplay duration, screen use) and biological material (fecal samples), data governance should be structured from the outset according to a principle of data minimization. At the collection stage, only variables strictly necessary to address the research question should be obtained. For potentially sensitive materials such as screen-time screenshots, participants should receive explicit masking instructions, and the upload interface should incorporate automated de-identification procedures (e.g., removal or cropping of notification bars, contact names, or message previews).

At the storage level, a separated key system is recommended: personally identifiable information (e.g., contact details) should be stored in a secure linkage file distinct from the analytical dataset, connected through anonymized study identifiers. Access controls should be role-based, and

audit logs should document data access and modification events. These measures reduce the risk of re-identification and support compliance with data protection standards.

To enhance reproducibility, supplementary materials should include a detailed variable dictionary, prespecified data cleaning rules, and a minimal working analysis script sufficient to reproduce core tables and models. If raw data cannot be publicly shared due to privacy constraints, a synthetic data generation script with clearly specified parameter ranges should be provided. The manuscript should explicitly document the data-generating rules and random seed strategy to ensure replicability of analytic outputs. In studies that initially demonstrate a framework using synthetic data, transparency in model specification, table structure, and reporting logic is essential so that readers can replicate analytical procedures without accessing identifiable participant information.

2. 2. Results

在便形方面，BSFS 均值随时长增加而下降，硬便比例与复合便秘样比例呈梯度上升。值得注意的是，Q4 组中 10.1–15 分钟区间的占比显著增大，提示长时长玩家更可能出现“延长坐厕”行为。化学子样本中，含水率下降与 pH 上移与硬便表型方向一致，而丁酸下降/氨上升则为讨论饮食结构与发酵模式变化提供了补充线索。

本节结果均为合成数据，用于展示常见表格组织与文字叙述方式。总体上，从 Q1 到 Q4，乙游周时长与“如厕刷手机评分”呈同步上升趋势，这与乙游碎片化使用更容易渗透到如厕场景的假设一致。与此同时，排便频次呈轻度下降，而单次如厕时长呈明显上升，提示‘减少排便次数 + 单次耗时更长’这一组合表型。

回归模型结果显示，在不纳入如厕刷手机评分时，乙游时长对如厕时长的估计更大；纳入后系数下降但仍显著，体现了“部分中介/部分解释”的结构。这种写法在真实论文中很常见：先给出总效应（total association），再逐步加入路径变量观察系数变化，从而为行为机制讨论提供量化支撑。

就效应量而言，OR 在合成结果中用于展示写作方式：当复合便秘样结局的基线发生率较低时，OR 会显得较大，因此真实研究应同时报告绝对风险差或预测概率，以避免读者仅凭 OR 产生夸大印象。同样，线性回归的分钟数变化应与分布信息（如分位数、区间占比）联动解读，才能判断其在生活意义上的重要性。

补充结果呈现建议：在真实研究写作中，除表格外可增加两类可视化以提升可读性。第一类为剂量-反应图：以乙游时长为横轴、如厕时长/BSFS 为纵轴，叠加限制性立方样条的拟合曲线与 95%置信带，用于检验是否存在阈值或非线性。第二类为风险画像图：把“高时长+短睡眠”“高时长+低步数”等组合暴露以分组柱状图或热力图展示复合结局的预测概率，使读者能直观看到交互项的含义。

另一个常见但容易被忽略的结果是“分布的形状变化”。例如，平均如厕时长的增加可能主要来自右尾拉长（少数人极端久坐），而非整体平移。对此，建议报告分位数（P50、P75、P90）或使用分位数回归，避免均值被极端值主导。同时可在附录中给出敏感性分析：剔除如厕时长>30 分钟或>60 分钟者后，主效应是否保持方向一致。

对于化学指标，建议把结果叙述限定在“方向一致性与探索性”，并避免在样本量较小或批次效应明显时做过强机制推断。一个更稳妥的写法是：先报告质量控制（CV、LOD、批次调整）与数据分布，再报告回归系数/趋势，并将其与便形/含水率的方向进行对照。若发现某一指标方向不一致，也应如实呈现并讨论可能原因（底物差异、保存偏差、个体菌群差异），而不是强行闭环。

With respect to stool form, the mean value on the Bristol Stool Form Scale (BSFS) decreases as weekly engagement duration increases, while the proportion of hard stool (Types 1–2) and the composite constipation-like indicator show a graded upward trend. Notably, the proportion of individuals within the 10.1–15 minute single-episode toileting interval increases substantially in the highest exposure quartile (Q4), suggesting that longer-duration players are more likely to exhibit prolonged toilet-seated behavior. Within the chemical subsample, reduced fecal water content and upward shifts in pH are directionally consistent with the hard-stool phenotype. Concurrent trends toward lower butyrate and higher ammonia provide supplementary clues relevant to dietary substrate availability and shifts in fermentation patterns.

All results in this section are based on synthetic data and are presented to illustrate common table structures and narrative conventions. Overall, from Q1 to Q4, weekly engagement duration increases in parallel with the toileting smartphone use score, consistent with the hypothesis that fragmented game usage more readily extends into toileting contexts. Concurrently, bowel movement frequency shows a mild decline, whereas single-episode toileting duration shows a clearer upward shift, yielding a combined phenotype characterized by fewer weekly bowel movements and longer time per episode.

Regression analyses demonstrate that when the toileting smartphone use score is excluded from the model, the estimated association between game duration and toileting duration is larger. Upon inclusion, the coefficient attenuates but remains statistically significant, consistent with a partial mediation or partial explanation structure. This sequential modeling approach—first presenting the total association, then incrementally adjusting for pathway variables—is common in empirical writing and provides quantitative support for behavioral mechanism discussions.

Regarding effect magnitude, the synthetic results use odds ratios to demonstrate reporting format. When the baseline prevalence of a composite constipation-like outcome is low, odds ratios may appear large despite modest absolute risk differences. Accordingly, real-world analyses should report adjusted predicted probabilities or absolute risk differences alongside odds ratios to prevent misinterpretation. Similarly, changes in minutes derived from linear regression should be interpreted jointly with distributional metrics (e.g., quantiles or interval proportions) to assess their practical relevance.

For enhanced clarity in empirical reporting, two types of visualization are recommended in addition to tables. First, dose–response plots, with weekly engagement duration on the x-axis and toileting duration or BSFS on the y-axis, overlaid with restricted cubic spline fits and 95% confidence bands, can evaluate potential nonlinearity or threshold effects. Second, risk profile visualizations—such as grouped bar charts or heat maps combining high engagement with short sleep or low step count—can depict predicted probabilities of composite outcomes, thereby clarifying interaction effects.

An often-overlooked result concerns changes in distributional shape. For example, an increase in mean toileting duration may primarily reflect right-tail elongation driven by a small subset of extreme prolonged sitters, rather than a uniform upward shift. To address this, reporting selected quantiles (e.g., P50, P75, P90) or applying quantile regression is advisable. Sensitivity analyses excluding individuals with toileting duration exceeding 30 or 60 minutes should be presented to assess robustness of the primary association.

For chemical indicators, interpretation should remain explicitly exploratory and limited to directional consistency. In small samples or in the presence of notable batch effects, strong mechanistic claims should be avoided. A prudent reporting sequence is to present quality-control metrics (e.g., coeffi-

cients of variation, limits of detection, batch adjustments), describe data distributions, then report regression coefficients or trend estimates and compare their direction with stool form and water content patterns. If certain indicators do not align directionally, these discrepancies should be transparently reported and discussed in light of plausible explanations, such as dietary substrate variation, storage-related bias, or inter-individual microbiota differences, rather than forcing interpretive closure.

Table 3a. Baseline characteristics by playtime quartile.

Quartile	n	Playtime (h/wk)	Age	BMI	Sleep (h/d)	Steps/d	Fiber (g/d)	Water (mL/d)	Phone score
Q1	370	2.37	24.39	21.88	6.98	7185	17.64	1689	2.86
Q2	370	4.54	24.13	22.37	6.93	7304	16.90	1715	3.54
Q3	370	6.93	24.21	22.52	6.93	7219	17.10	1712	4.05
Q4	370	11.96	24.15	22.22	6.96	7280	17.41	1661	5.37

Table 3b. Bowel outcomes by playtime quartile.

Quartile	BM/week	Toilet duration (min)	Stool length (cm)	BSFS	Hard stool (%)	Composite (%)
Q1	7.00	6.75	10.54	3.91	8.6	8.6
Q2	6.82	7.14	10.51	3.68	14.3	14.3
Q3	6.89	7.72	10.18	3.59	15.9	15.9
Q4	6.55	9.19	9.77	3.33	21.1	25.4

Table 4. Toilet duration distribution by playtime quartile.

Quartile	≤5 min	5.1–10 min	10.1–15 min
Q1	13.0%	84.6%	2.4%
Q2	8.1%	89.5%	2.4%
Q3	4.6%	90.0%	5.4%
Q4	0.0%	69.5%	30.5%

Table 5. BSFS category distribution by playtime quartile.

Quartile	Normal (3–4)	Hard (1–2)	Soft/loose (5–7)
Q1	63.2%	8.6%	28.1%
Q2	65.4%	14.3%	20.3%
Q3	63.8%	15.9%	20.3%
Q4	65.9%	21.1%	13.0%

Table 6. Prevalence of selected otome lexicon items by playtime quartile.

Quartile	上头	我推	吃谷	保底	爆肝	回血	梦女	抽卡
Q1	18.9%	29.7%	13.5%	13.2%	17.6%	20.8%	15.9%	33.0%
Q2	27.8%	35.7%	19.7%	26.2%	21.1%	30.0%	22.2%	38.4%
Q3	40.3%	45.9%	26.5%	30.3%	34.6%	34.3%	28.6%	48.4%
Q4	67.0%	70.5%	51.6%	54.3%	59.2%	58.9%	50.0%	66.5%

Table 7. Stool chemistry markers by playtime quartile.

Quartile	n	Water (%)	pH	Acetate	Propionate	Butyrate	BCFA	Ammonia
Q1	159	75.02 (73.79–76.09)	6.66 (6.47–6.85)	46.00 (39.68–51.42)	15.86 (13.69–18.03)	12.13 (10.00–13.85)	5.38 (4.62–6.28)	16.19 (13.42–19.39)
Q2	158	75.09 (74.04–75.95)	6.72 (6.52–6.90)	45.24 (39.03–50.83)	15.60 (12.92–17.88)	12.11 (10.34–13.79)	5.75 (4.95–6.45)	18.05 (14.86–20.91)
Q3	158	74.68 (73.46–75.54)	6.77 (6.59–6.98)	44.65 (39.38–49.60)	15.19 (13.43–17.96)	11.18 (9.06–12.99)	5.65 (4.85–6.49)	19.19 (16.23–22.18)
Q4	158	74.41 (73.15–75.48)	6.81 (6.67–7.04)	43.86 (37.20–49.95)	14.96 (13.03–17.27)	11.08 (8.99–13.77)	5.98 (5.04–6.98)	20.49 (17.54–22.98)

Table 8. Multivariable regression results.

Outcome / Predictor	Effect	95% CI	p	n
Toilet duration, playtime per +10 h/wk (without phone score)	β=2.531	2.321–2.741	<0.001	1480
Toilet duration, playtime per +10 h/wk (adjusted incl. phone score)	β=1.618	1.389–1.847	<0.001	1480
Toilet duration, phone score per +1	β=0.362	0.317–0.406	<0.001	1480
BM/week, playtime per +10 h/wk (adjusted)	β=-0.389	-0.531–-0.247	<0.001	1480
Hard stool, playtime per +10 h/wk (adjusted)	OR=2.643	1.724–4.052	<0.001	1480
Composite constipation-like, playtime per +10 h/wk (adjusted)	OR=3.768	2.482–5.721	<0.001	1480

Table 9. Chemistry regression models.

Chemistry outcome (per +10 h/wk)	β	95% CI	p	n
Fecal water (%)	-0.662	-1.039–-0.285	<0.001	633
Fecal pH	0.139	0.080–0.198	<0.001	633
Butyrate (μmol/g)	-0.695	-1.370–-0.021	0.043	633
Ammonia (mmol/kg)	3.347	2.350–4.344	<0.001	633

3. Discussion

3.1 主要发现与解释（Principal findings）

本合成数据研究呈现出一个内部一致的结果模式：乙游时长越高，单次排便时长越长、BSFS 分值越低、硬便比例与复合便秘样比例越高。在化学子样本中，含水率下降与 pH 升高的方向与“偏干、排出更慢”的表型相匹配；丁酸下降与氨升高则提供了一个可讨论的代谢层解释：当饮食纤维不足、久坐和作息紊乱共存时，发酵底物与代谢产物谱可能出现轻度变化。需要强调的是，这些解释在真实研究中必须依赖更完整的膳食结构与样本处理信息验证，本稿仅用于展示如何将行为结果与化学指标写成一个逻辑闭环。

化学指标的讨论需要谨慎。丁酸等 SCFAs 与结肠健康的关系在综述中被广泛讨论，但粪便中测得的浓度同时受产生、吸收与排出影响；pH 也会受短链脂肪酸谱与蛋白发酵产物共同影响。因此，在真实研究写作中，应把化学结果定位为“方向一致性”而非因果证据，并把样本处理、检测批次与饮食底物信息作为解释前提。

将行为表型与机制线索并置，有助于把讨论从‘游戏本身’转向‘与游戏共现的生活方式结构’。既有公共健康资料强调纤维、饮水与活

动量对便秘的重要性，而系统综述也提示中高水平身体活动与较低便秘风险相关；因此，在解释乙游时长效应时，更合理的表述是：乙游高时长可能作为一个“久坐+节律扰动”的综合代理，其效应大小与个体的饮食、饮水与运动背景密切相关。

从实践角度看，若真实研究复现类似关联，其公共健康含义更可能落在“习惯改造”而非“娱乐否定”。例如，增加日常步行与站立时长、在活动周特别注意饮水与膳食纤维、以及给如厕场景设置明确的结束条件（不把厕所当作刷屏空间），这些都是成本较低且可操作的策略。当然，任何具体干预建议都应建立在真实数据与个体差异评估之上，本模板仅提供写作讨论框架。

This synthetic study demonstrates an internally coherent pattern: higher otome playtime is associated with longer toilet duration, lower BSFS scores, and higher prevalence of hard stool and the composite constipation-like outcome. In the chemistry subsample, lower fecal water and higher pH align with a drier, slower-evacuation phenotype, while lower butyrate and higher ammonia provide a discussable metabolic-layer narrative: when low fiber intake co-occurs with sedentary structure and disrupted routines, substrate availability and metabolite profiles may shift modestly. Crucially, these interpretations would require validation with richer dietary data and biospecimen handling metadata in real studies; here, they serve to demonstrate how behavioral and chemistry results can be written into a coherent argument.

3.2 ‘如厕刷手机’的统计与行为含义（Phone-on-toilet pathway）

在模型中加入“如厕刷手机评分”后，乙游时长对排便时长的系数下降但仍显著，说明该路径具有“部分解释”作用。这一结果在真实场景中是可理解的：排便时长既可以反映排出困难，也可以反映行为停留。对于乙游玩家而言，如厕时继续浏览社区、查看活动攻略、回收体力（俗称“回血”）或推进剧情，是一种常见的碎片化使用形态；因此，若不测量这一行为，研究很可能把‘厕所里的手机停留’误判为‘排便困难’。在真实研究写作中，建议将排便用力程度（straining）与是否排净（incomplete evacuation）作为补充终点，以提高对“生理困难”的分辨率。

‘如厕刷手机’路径的价值在于把一个常见但易被忽略的行为纳入了可量化模型。近期关于“厕所用手机与痔疮风险”的研究提示，手机会显著延长坐厕时间，而延长坐厕可能带来局部压力增加等风险。即便本研究终点并非痔疮，这类证据仍提示：在公共健康建议中，与其单纯强调‘少玩游戏’，不如更可操作地建议‘如厕控制坐位停留、把手机留在厕所外’，以减少把厕所变成“第二个刷屏场景”的机会。

在更偏应用的讨论里，还可以把“如厕场景”作为行为干预的最小单元：一方面设置明确的时长上限（例如完成排便后即离开坐厕），另一方面用环境提示替代意志力（把手机放在看不见的地方、在卫生间门口放置水杯提醒补水、在活动周额外安排短步行）。这些措施并不依赖对游戏的价值判断，而是针对高频、可重复、易被忽视的小习惯。若未来实证研究能够验证这些微干预与排便指标的改善相关，将更有可能转化为可推广的健康建议。

After inclusion of the toileting smartphone use score in the regression model, the coefficient linking otome game duration to defecation duration attenuates but remains statistically significant. This pattern is consistent with partial explanation rather than full mediation. Conceptually, this finding is plausible. Single-episode toileting duration reflects both physiological evacuation difficulty and behavioral prolongation. Among otome players, continued engagement during toileting—such as browsing community forums, checking event guides, recovering in-game stamina, or progressing through narrative content—represents a common form of fragmented use. In the absence of explicit measurement of this behavior, prolonged screen-related sitting may be misclassified as physiological defecatory difficulty. In empirical research, inclusion of additional endpoints such as perceived straining and incomplete evacuation is advisable to improve discrimination between behavioral delay and genuine physiological impairment.

The analytical value of the “toileting smartphone use” pathway lies in formalizing a high-frequency but often overlooked behavior within a quantifiable model. Recent studies examining smartphone use in restroom contexts, particularly in relation to hemorrhoidal risk, suggest that device use substantially extends toilet-seated duration and may increase local pressure exposure. Even though hemorrhoids are not the primary endpoint in the present framework, this literature reinforces the broader inference that prolonged seated duration is behaviorally modifiable. From a public health perspective, the implication is that interventions may be more actionable if framed not as generalized “reduce gaming” messages, but as context-specific recommendations such as limiting seated time during toileting or keeping smartphones outside the restroom to prevent conversion of a brief physiological event into an extended screen-based episode

In a more applied discussion, the toileting context can be conceptualized as a minimal behavioral intervention unit. Practical strategies may include establishing a clear upper bound on seated duration (e.g., leaving the toilet promptly after evacuation), substituting environmental cues for reliance on willpower (e.g., placing the smartphone out of reach, positioning a water bottle outside the bathroom as a hydration reminder, scheduling brief walking breaks during intensive event weeks). These measures do not depend on normative judgments about gaming itself; rather, they target frequent, repeatable, and often unnoticed micro-behaviors. If future empirical studies confirm that such micro-interventions are associated with measurable improvements in bowel-related indicators, they may offer a pragmatic pathway toward scalable health recommendations.

3.3 乙游“梗语”变量的科学化用法（From memes to measurable constructs）

本稿将“梗语”作为可量化的文化参与指标，而不是背景彩蛋。其核心价值在于：它能将同样的‘时长’区分为不同的参与结构。例如，同为 10 小时/周，有人可能主要是挂机刷日常；有人则在活动周集中“爆肝”，并伴随高强度社交讨论与“吃谷”。梗语指数可以用于三种分析：其一，作为暴露的补充（替代或补充时长）；其二，作为效应修饰变量，检验高沉浸文化参与者是否更容易出现如厕手机使用与排便时长延长；其三，作为数据质量的交叉校验（极高时长但零梗语可能提示测量偏差）。

把“梗语”纳入量表并不是为了增加趣味性，而是为了把亚文化参与从不可见变为可测量。更高的梗语沉浸往往意味着更高的社群参与、更高的情绪唤起与更强的事件周投入，这些因素可能改变睡眠、饮食与活动，从而对排便产生间接影响。真实研究中，可把梗语指数作为效应修饰变量，检验‘高时长但低沉浸’与‘高时长且高沉浸’是否存在不同的风险画像。

In this framework, subcultural “meme vocabulary” is operationalized as a quantifiable indicator of cultural participation rather than treated as incidental background detail. Its analytical value lies in differentiating engagement structures that may otherwise appear equivalent in terms of total duration. For example, two individuals reporting 10 hours per week of gameplay may differ substantially: one may primarily engage in low-intensity routine tasks or idle modes, whereas another may concentrate exposure during event weeks with high-intensity participation, concurrent social discussion, and merchandise-related behaviors. A meme immersion index enables this structural differentiation.

The index may be incorporated into analysis in at least three ways. First, it can function as a complementary exposure metric, either supplementing or partially substituting for total duration. Second, it may serve as an effect modifier, allowing examination of whether individuals with higher cultural immersion are more likely to exhibit extended toileting smartphone use or prolonged defecation duration at comparable exposure levels. Third, it can provide a form of internal data validation: discordant patterns (e.g., extremely high reported duration accompanied by zero immersion indicators) may signal potential measurement error or misclassification.

The inclusion of meme vocabulary is therefore not intended to add narrative color but to render subcultural participation measurable. Higher immersion scores plausibly correspond to greater community involvement, heightened emotional arousal, and more intensive event-week engagement. These factors may indirectly influence sleep patterns, dietary behavior, and physical activity, thereby affecting bowel-related outcomes. In empirical applications, stratified or interaction analyses can assess whether "high duration–low immersion" and "high duration–high immersion" groups exhibit distinct risk profiles, thereby refining interpretation beyond simple dose–response models based solely on time exposure.

3.4 局限性与真实研究建议（Limitations and recommendations）

局限性首先来自研究性质：所有数据为合成结果，不能用于实证推断。其次，排便行为强依赖自评，真实研究需要通过重复测量与客观指标校验降低误差。对化学子样本而言，采样时间窗、保存温度与冻融次数都会影响读数，必须记录并作为质控信息进入分析。未来若开展真实研究，建议采用：（1）更精细的饮食测量（可发酵纤维、总能量与蛋白比例）；（2）更细的久坐测量（加速度计或手机健康数据）；（3）分层/交互模型检验活动周与睡眠不足的叠加效应；（4）在统计上优先采用个体内对照（within-person）分析，以削弱个体差异造成的混杂。

第三，因果表述应克制：在没有随机化或外生冲击设计的情况下，更稳妥的结论是"在控制可测混杂后仍存在稳定关联"。若要更接近因果推断，可考虑把游戏活动开启视为自然实验、或随机分配'如厕不带手机'等微干预来检验排便时长是否改变。

第二，测量误差是排便研究的核心挑战。自报 BSFS 与时长易受记忆偏差与社会期许影响，且不同人对"开始/结束"的界定不同。真实研究应通过重复测量、明确的填报指引与必要的培训材料（图示、示例）来降低误差，并在附录报告一致性检查结果。

第一，混杂与反向因果需要被明确写入局限性：压力、焦虑或睡眠问题可能同时增加游戏作为应对方式的使用，并直接影响肠道功能。若无法充分测量这些因素，乙游时长的系数可能部分反映心理与作息状态，而非游戏暴露本身。

第七，报告规范方面，可参考 STROBE 等观察性研究报告指南来组织方法与结果，尤其是在样本流程图、缺失值处理、敏感性分析与混杂控制策略的描述上。对读者而言，'写得清楚'往往比'结果更显著'更能提升论文的可接受度。

第六，化学子样本的前处理与检测策略会直接影响可解释性。建议在真实研究中补充：采样工具与防污染步骤、保存缓冲液与否、冷链时长、冻融次数、检测平台（GC-MS/LC-MS 等）与定量方法（内标、校准曲线）。这些信息写清楚后，即便效应较小，也更容易被读者视为可信的探索性结果。

第五，饮食测量的"关键缺口"往往在于可发酵纤维与零食结构。即便总纤维克数相近，来源不同（全谷物、豆类、蔬果、抗性淀粉）对发酵底物与 SCFAs 谱的影响也可能不同。若论文希望讨论丁酸变化，更应在方法中说明纤维来源的测量方式，并在讨论中承认'底物信息不足'会限制机制解释。

第四，代表性与可推广性需要谨慎。乙游玩家群体在年龄、性别比例、学习/工作压力与社交媒体使用上可能与一般人群不同，若样本来自单一平台或单一地区，外推到更广泛人群时应明确限制。真实研究可通过分层招募（不同职业/学校、不同地区）与加权分析缓解偏差。

第十，在写作表达上，建议把讨论拆成"行为层""生理层""方法层"三条线：行为层解释久坐、睡眠与厕所手机使用；生理层解释水分、便形与代谢指标的方向一致性；方法层强调合成数据性质、测量误差与因果边界。这样的结构能减少读者把论文理解为道德评判或夸大因果的风险。

第九，若研究者担心自报时长的社会期许偏差，可采用"区间报告 + 置信度评分"的方式：让参与者选择时长区间（如 0–2、2–5、5–10、>10 小时/周）并报告自信程度，再在模型中进行误差敏感性分析。对青少年样本尤其重要，因为他们可能因家长/学校规范而低报娱乐时长。

第八，未来研究可以采用生态瞬时评估（EMA）与被动感知结合的方式：在关键时段（睡前、如厕后、活动周开启时）进行短问卷推送，同时利用设备传感数据记录久坐时间、步数与屏幕使用。这样能把"何时玩、在哪里玩、在什么情绪下玩"与"何时排便、排便是否困难"更精细地对齐，从而提高机制推断的可信度。

Several limitations warrant explicit acknowledgment. First, all results presented are based on synthetic data and therefore cannot support empirical inference. The purpose of the analysis is structural demonstration rather than evidentiary conclusion. Any real-world interpretation must await validation using observational or experimental datasets.

Second, measurement error represents a central challenge in bowel-related research. Defecation frequency, stool form (e.g., via the Bristol Stool Form Scale), and toileting duration are typically self-reported and may be influenced by recall bias, social desirability, and heterogeneous interpretations of "start" and "end." Real-world studies should mitigate these issues through repeated measurements, explicit reporting instructions, and standardized visual materials. Consistency checks and reporting of reliability metrics in appendices are advisable.

Third, residual confounding and reverse causation must be explicitly addressed. Psychological stress, anxiety, or sleep disturbance may simultaneously increase reliance on gaming as a coping strategy and independently affect gastrointestinal function. Incomplete measurement of these factors may lead to partial misattribution of associations to gaming duration rather than underlying psychosocial states. In the absence of randomization or exogenous shocks, conclusions should remain framed as stable associations after adjustment for measurable confounders rather than causal effects.

Fourth, representativeness and generalizability require caution. Otome game players may differ from the general population in age distribution, gender composition, stress exposure, and social media intensity. Samples drawn from a single platform or geographic region further limit external validity. Stratified recruitment across occupations, schools, and regions, combined with weighting or sensitivity analyses, can partially address these concerns.

Fifth, dietary measurement frequently contains critical gaps, particularly regarding fermentable fiber sources and snack structure. Equivalent total fiber intake may derive from qualitatively different substrates (e.g., whole grains, legumes, fruits and vegetables, resistant starch), each with distinct implications for microbial fermentation and short-chain fatty acid profiles. If discussion extends to butyrate or related metabolites, methodological sections should clearly specify how fiber sources are

assessed, and discussion sections should acknowledge substrate uncertainty as a constraint on mechanistic inference.

Sixth, pre-analytical handling and detection strategies in the chemical subsample directly affect interpretability. Transparent reporting of sampling tools, contamination prevention procedures, use of stabilizing buffers, cold-chain duration, freeze–thaw cycles, analytical platforms (e.g., GC–MS or LC–MS), and quantification methods (internal standards, calibration curves) is essential. Even modest effect sizes gain credibility when methodological rigor is clearly documented.

Seventh, reporting standards should align with established observational guidelines such as STROBE Statement. Clear description of participant flow, missing-data handling, sensitivity analyses, and confounder control strategies enhances interpretability and transparency. Methodological clarity often contributes more to manuscript credibility than marginal increases in statistical significance.

Eighth, future research may integrate ecological momentary assessment (EMA) with passive sensing. Timed micro-surveys during critical periods (e.g., before sleep, after toileting, at event-week onset) combined with device-recorded sedentary time, step counts, and screen use can align temporal exposure patterns more precisely with bowel-related outcomes, strengthening mechanistic inference.

Ninth, to address potential social desirability bias in self-reported gaming duration—particularly among adolescents—an interval-report format combined with confidence ratings may be employed. Subsequent measurement-error sensitivity analyses can assess robustness of exposure estimates.

Tenth, for interpretive clarity, discussion sections may be organized along three parallel axes: a behavioral layer (sedentary patterns, sleep timing, toileting smartphone use), a physiological layer (hydration status, stool form, metabolic indicators and directional coherence), and a methodological layer (synthetic data nature, measurement limitations, causal boundaries). Such structuring reduces the risk that readers misinterpret the study as moral evaluation or overstate causal implications.

Future empirical research would benefit from: (1) more granular dietary assessment, including fermentable fiber, total energy, and protein proportions; (2) objective sedentary measurement via accelerometry or device health logs; (3) stratified or interaction modeling to test combined effects of event-week intensity and sleep deprivation; and (4) greater emphasis on within-person analytic strategies to reduce confounding by stable individual differences.

3.5 机制层解释与替代假说（Mechanistic interpretation and alternative hypotheses）

在机制讨论中，建议采用“由近及远”的层级：先解释最接近观测指标的行为机制，再讨论可能的生理机制，最后再触及化学指标的推测。对本主题而言，最直接的机制是‘注意力占用导致的如厕停留’，它会把如厕时长抬高而不必然伴随排便困难；其次是‘作息与活动结构’改变，它可能通过减少步行、缩短睡眠、推迟如厕等方式影响排便节律；最后才是‘底物与发酵’层面的可能变化（纤维不足、蛋白发酵增强、pH 上移、氨上升、丁酸下降）。这样的写法既能解释为什么纳入如厕手机使用后系数会衰减，也能避免把化学结果写成过度确定的因果链条。

For mechanism discussion, a ‘near-to-far’ hierarchy is useful: start with behavioral mechanisms closest to the observed measures, then consider plausible physiological mechanisms, and only then discuss chemistry-level speculation. Here, the most proximate mechanism is attention-driven toilet sitting, which can inflate duration without true evacuation difficulty. Next is altered routine and activity structure, influencing bowel rhythms via reduced walking, shorter sleep, delayed toileting, and irregular hydration. Only after that should the manuscript discuss substrate/fermentation shifts (lower fiber, more protein fermentation, higher pH, higher ammonia, lower butyrate). This ordering explains why coefficients attenuate after adjusting for phone-on-toilet behavior and avoids presenting chemistry findings as a deterministic causal chain.

替代假说需要被主动写入讨论以约束因果表述。第一类替代解释是饮食驱动：活动周或沉浸剧情可能增加零食与咖啡因摄入、减少蔬果与全谷，从而改变含水率与 SCFAs。第二类是压力驱动：考试周/工作高峰既可能增加游戏作为应对方式，也可能通过自主神经与睡眠影响排便。第三类是稳定特质：有些人本就习惯在厕所久坐刷手机，导致如厕时长与任何屏幕暴露都相关。写作上，建议用三段式回应：说明已控制哪些变量、还可能遗漏哪些变量、以及未来如何用设计补齐（个体内比较、自然实验、随机化微干预）。

Alternative hypotheses should be explicitly discussed to constrain causal language. One is diet-driven confounding: event weeks or immersive stories may increase snacks and caffeine while reducing fruits/vegetables and whole grains, shifting fecal water and SCFAs. Another is stress-driven confounding: exam/work peaks can increase gaming as coping and influence bowel function via autonomic pathways and sleep. A third is stable traits: some individuals habitually sit and scroll on toilets, making toilet duration correlate with many forms of screen exposure. A practical reporting template is a three-part response: what has been controlled, what may still be unmeasured, and how future designs can address it (within-person comparisons, natural experiments, randomized micro-interventions).

3.6 可操作建议与产品侧启示（Actionable guidance and design implications）

若研究目标是提出可执行的健康建议，最可行的切入点往往不是“减少兴趣活动”，而是“重塑高频场景”。如厕场景具有三点优势：高频、可定义、可被环境改造。具体建议可包括：设定如厕坐位上限（如 5–8 分钟）、把手机放在卫生间外或使用“离开时提醒”、排便后立即起身并补水、活动周额外安排短步行以抵消久坐。与其把问题归因于某种游戏，不如把它写成“屏幕渗透到生活微场景后带来的节律风险”，这样更容易被不同受众接受。

If the goal is actionable guidance, the most feasible entry point is often not ‘reducing an interest activity’ but ‘reshaping high-frequency contexts.’ Toileting has three advantages as an intervention unit: it is frequent, well-defined, and modifiable by environment. Concrete suggestions include setting a sitting-time cap (e.g., 5–8 minutes), keeping phones outside bathrooms or using ‘leave’ reminders, standing up promptly after evacuation and drinking water, and adding short walks during event weeks to offset sedentary spikes. Framing the issue as ‘screen-time seepage into micro-contexts and rhythm risks’ is generally more acceptable and transferable than attributing the problem to a specific game genre.

对产品侧而言，研究也可提出不带道德评判的“减害设计”思路。例如，在活动周高峰时段，应用内外的提醒机制可能鼓励更规律的休息：番茄钟式的休息提示、在连续使用达到一定阈值后展示“起身走动/补水”提示、以及把“挂机/自动战斗”从必须常驻前台改为更友好的后台策略（若游戏机制允许）。当然，这类建议需要谨慎：研究者应强调其目标是降低久坐与作息扰动，而不是干预玩家偏好或商业模式。

From a product perspective, the manuscript can also propose value-neutral ‘harm-reduction’ design ideas. During event-week peaks, reminders can encourage regular breaks: pomodoro-style prompts, hydration/movement nudges after prolonged continuous use, and (when mechanics allow) shifting auto-battle or idle routines away from requiring constant foreground attention. Such suggestions should be framed care-

fully: the goal is to reduce sedentary time and rhythm disruption rather than to police preferences or dictate business models.

4. Conclusion

在模拟 14 天日记研究结构下，乙游周游玩时长与排便时长延长、便形偏硬及化学指标方向一致的轻度变化相关。本文提供可替换为真实数据的双语论文模板。

在未来的实证研究中，建议优先采用短周期重复测量与个体内分析，并把饮食底物、睡眠节律与久坐结构作为关键协变量或交互项处理。这样可以把“乙游时长”从单一标签转化为更具公共健康意义的行为画像。

综合而言，本稿的贡献在于提供了一套可直接迁移到真实数据的双语写作与分析框架：从暴露刻画（时长、活动周、文化沉浸）到路径变量（如厕刷手机）再到多维结局（频次、时长、便形与化学指标），形成一条可解释的论证链。

最后，建议将问卷条目、日记表单字段、清洗规则与分析代码（在不包含任何个人隐私的前提下）以补充材料形式公开或可申请获取。这样做不仅提升可复现性，也能让后续研究者更容易把该框架替换为真实数据并进行跨样本验证。

若将本框架用于真实数据替换，建议在摘要与讨论中明确区分“统计关联”与“健康风险”，并在结尾给出对读者友好的可操作建议（例如规律作息、增加活动、控制如厕刷手机），同时强调个体差异与医疗咨询边界。这种写法既能保持论文的公共健康价值，也能避免把研究主题简化为对特定兴趣群体的标签化。

补充写作建议：若把本模板用于真实研究投稿，可在补充材料中提供日记表单截图、变量字典、以及最小可运行的统计脚本（R/Python 均可），并在正文中用一张概念框架图展示“乙游时长—久坐/睡眠/如厕手机—排便表型—化学指标”的路径结构。对于读者而言，框架图能显著降低理解成本，也能使论文从“现象描述”更自然地过渡到“可检验的机制假设”。

同时需要提醒：排便表型具有高度个体差异，短期波动并不等同于疾病。若出现持续便秘、便血、体重下降或明显腹痛等警示信号，应优先寻求医疗评估。本文的写作框架仅用于研究方法演示，不构成诊断或治疗建议。具体处理请以医生意见为准。

Under a simulated 14-day diary design, greater weekly otome game engagement is associated with prolonged single-episode toileting duration, harder stool patterns, and directionally consistent mild shifts in fecal chemical indicators. The present manuscript provides a bilingual template that can be directly adapted to empirical datasets.

For future real-world studies, priority should be given to short-cycle repeated measurement designs and within-person analytical strategies. Dietary substrates, sleep rhythm, and sedentary structure should be treated as key covariates or interaction terms rather than peripheral adjustments. This approach allows “weekly gaming duration” to move beyond a single exposure label and instead be reframed as a structured behavioral profile with clearer public health relevance.

Overall, the principal contribution of this work lies in offering a transferable analytic and writing framework: from exposure characterization (duration, event-week intensity, cultural immersion), to pathway variables (toileting smartphone use), to multidimensional outcomes (frequency, duration, stool form, and chemical indices), forming a coherent explanatory chain. The framework is intentionally modular so that synthetic inputs can be replaced with real-world data without altering the conceptual architecture.

To enhance reproducibility, survey instruments, diary field structures, data cleaning rules, and minimal executable statistical scripts (excluding any personal identifiers) should be made available as supplementary materials or upon request. Such transparency not only strengthens methodological credibility but also facilitates cross-sample validation and independent replication.

If applied to empirical data, abstracts and discussions should clearly distinguish statistical association from clinical risk. Concluding sections may include pragmatic, reader-oriented recommendations—such as maintaining regular sleep schedules, increasing daily movement, and limiting smartphone use during toileting—while emphasizing inter-individual variability and the boundaries of medical advice. This framing preserves public health relevance without reducing the topic to a normative judgment about a particular interest group.

For manuscript preparation, inclusion of a conceptual framework diagram illustrating the pathway structure— “gaming duration → sedentary/sleep/toileting smartphone behavior → bowel phenotype → chemical indicators” —can substantially reduce interpretive burden for readers and support a transition from descriptive findings to testable mechanistic hypotheses.

Finally, bowel patterns exhibit substantial inter-individual variability, and short-term fluctuations do not equate to disease. Persistent constipation, rectal bleeding, unexplained weight loss, or significant abdominal pain warrant medical evaluation. The framework presented here is intended solely for methodological illustration and does not constitute diagnostic or therapeutic guidance. Clinical decisions should be based on professional medical consultation.

References

NIDDK. Definition & Facts for Constipation. https://www.niddk.nih.gov/health-information/digestive-diseases/constipation/definition-facts

Mayo Clinic. Constipation: Symptoms and causes. https://www.mayoclinic.org/diseases-conditions/constipation/symptoms-causes/syc-20354253

Blake MR, Raker JM, Whelan K. Validity and reliability of the Bristol Stool Form Scale (BSFS). Aliment Pharmacol Ther. 2016; PMID: 27492648.

Ramprasad C, et al. Smartphone use on the toilet and the risk of hemorrhoids. PLOS ONE. 2025. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0329983

Xiong YJ, et al. Joint association of sitting time and sleep duration with constipation (NHANES). Frontiers in Nutrition. 2025. https://www.frontiersin.org/journals/nutrition/articles/10.3389/fnut.2025.1548455/full

Ohkuma T, et al. Sleep duration and its association with constipation. PLOS ONE. 2024; e0302430.

Cui J, et al. Physical activity and constipation: A systematic review. (PMC) 2024.

Yamamura R, et al. Intestinal and fecal pH in human health. Frontiers in Microbiomes. 2023.

Liu H, et al. Butyrate: A double-edged sword for health? (PMC) 2018.

Rome Foundation. Rome IV Criteria: Functional Constipation. 2016 (web).

5. Supplementary Extended Sections (Bilingual)

S1. 扩展背景：乙游时间结构、情绪经济与生活方式耦合（Extended background）

为更精细地解释“乙游时长”这一暴露，需要把它拆成可观测的时间单元与场景单元。对多数乙游而言，日常循环由“固定刷新任务—限时活动—剧情更新”三部分组成：固定任务提供低门槛的每日登录惯性；限时活动在短窗口内制造强烈的时间压力与 FOMO（错过恐惧）；剧情更新与角色陪伴则提供持续的情绪回报。三者叠加后形成一种独特的时间结构：既有稳定的日常底座，又有活动周的高峰脉冲，还会把碎片时间（通勤、课间、睡前、如厕）系统性占满。

To interpret otome ‘playtime’ more precisely, it helps to decompose it into observable time units and contexts. For most otome games, engage-

ment follows a three-layer cycle: daily refresh tasks create a low-threshold login habit; limited-time events generate concentrated time pressure and FOMO; and story updates plus character companionship deliver sustained affective rewards. Together, these layers produce a distinctive time structure: a stable daily baseline, high-intensity event-week pulses, and systematic occupation of micro-moments (commuting, breaks, bedtime, and toileting).

从“情绪经济”的角度，乙游并不只是消磨时间，而常被玩家用作情绪调节工具：在压力上升时提供即时的可控反馈（刷任务、抽卡、推进剧情），在孤独或疲惫时提供拟亲密的陪伴体验。问题在于，情绪调节往往与生理节律同时被牵动：为了赶活动而晚睡、为了连胜或保底而延长连续使用、为了不被打断而减少起身活动，都会让睡眠与运动在短期内被压缩。因而，“高时长”更像是一组共发生的生活方式变化，而非单一的娱乐偏好。

From an ‘affective economy’ perspective, otome play is often used as emotion regulation rather than pure time-killing: it offers controllable, immediate feedback (tasks, gacha, story progression) during stress and a companionship-like experience during loneliness or fatigue. The issue is that emotion regulation can tug on physiological rhythms at the same time. Staying up late to chase events, extending continuous use to reach a pity threshold, or reducing movement to avoid interruptions can compress sleep and activity in the short run. Thus, ‘high playtime’ often reflects a cluster of co-occurring lifestyle shifts rather than a single entertainment preference.

乙游社群的梗语系统可进一步揭示“参与结构”。例如，“爆肝”强调的是活动周的集中投入；“保底/吃井”强调的是在不确定性驱动下的反复尝试与持续在线；“吃谷”提示线上娱乐向线下消费外溢；“回血”提示其被用作情绪恢复手段。研究上，把这些词汇转化为条目化频次与场景（例如：是否在如厕/睡前使用时更频繁出现“爆肝”）可以帮助区分‘同样时长但不同节律’的个体，从而提升暴露刻画的解释力。

The community lexicon further reveals engagement structure. For example, ‘baogan’ emphasizes concentrated event-week investment; ‘baodi/chijing’ reflect repeated attempts and persistent online presence under uncertainty; ‘chigu’ indicates spillover from online play to offline consumption; and ‘huixue’ signals emotional replenishment. Converting these terms into itemized frequencies and contexts (e.g., whether ‘baogan’ language is more common during toileting or bedtime use) can help differentiate individuals with the same total hours but different rhythms, improving exposure interpretability.

最后，场景单元的拆解对排便研究尤其关键。排便相关行为发生在高度具体的空间与时间：厕所、早晨起床后、餐后等。若乙游使用更倾向于渗透进这些固定场景（例如如厕时刷社区、睡前补剧情），那么它对排便的影响路径更可能通过“延迟如厕—延长坐厕—饮水减少”这类微行为体现，而不是通过宏观的总时长直接体现。这也是为什么本框架强调记录如厕手机使用与便意延迟等变量。

Finally, context decomposition is particularly important for bowel research because toileting occurs in highly specific spaces and times—bathrooms, after waking, after meals. If otome use preferentially penetrates these fixed contexts (scrolling community feeds on the toilet, reading story chapters before bed), pathways are more likely to operate through micro-behaviors such as urge delay, prolonged sitting, and reduced hydration, rather than through total hours alone. This is why the framework explicitly measures phone use on the toilet and urge-delay variables.

S2. 扩展方法：日记设计、依从性与偏倚控制（Extended methods considerations）

日记研究的关键在于把“想测的变量”翻译成参与者能稳定填写的字段。建议把每日记录拆成两层：事件层（每次排便事件：时间、BSFS、用力程度、是否排尽、如厕总时长、如厕手机使用时长）与日汇总层（当日总饮水、纤维相关摄入、总步数、睡眠起止、压力评分）。事件层尽量短、当场填写；日汇总层可在睡前一次填写。这样既能减少回忆偏差，也能提高字段之间的逻辑一致性。

The key to diary studies is translating target constructs into fields participants can reliably complete. A practical design uses two layers: an event layer for each bowel movement (time, BSFS, straining, incomplete evacuation, total toilet duration, phone-use duration on the toilet) and a daily summary layer (total water intake, fiber-relevant intake, steps, sleep start/end, stress rating). Event entries should be brief and completed close to the event; daily summaries can be completed once before bed. This reduces recall bias and improves internal consistency across fields.

依从性管理可采用“温和但持续”的策略：固定时间推送提醒（起床后、晚饭后、睡前）、提供一页图示的 BSFS 速查卡、以及允许在 24 小时内补填但标记为补填。研究者应预先定义完成率阈值（例如≥80%）并说明处理规则（完全案例或多重插补）。对关键字段（如排便次数、如厕时长），可设置逻辑校验：若当天报告排便 0 次，则不应出现 BSFS 与用力评分；若如厕总时长极端（如>60 分钟），提示参与者确认并记录原因（如刷手机、腹泻、便秘、身体不适）。

Compliance can be supported by ‘gentle but persistent’ mechanisms: scheduled reminders (after waking, after dinner, before bed), a one-page BSFS visual card, and allowing backfilling within 24 hours while flagging entries as backfilled. Predefine completion thresholds (e.g., ≥80%) and handling rules (complete-case vs multiple imputation). Implement logic checks: if BM=0, BSFS/straining should not appear; if toilet duration is extreme (e.g., >60 minutes), prompt confirmation and capture reasons (scrolling, diarrhea, constipation, discomfort).

暴露测量方面，自报时长可与设备屏幕使用时长形成三角校验。实际操作不必收集完整截图原图：可让参与者填写系统统计中的数值，并上传遮挡后的局部截图作为抽查验证。对多游戏玩家，建议分别记录每款乙游的时长并求和，同时记录“活动周强度”与“连续使用最长时段”，以区分总量与峰值。若研究对象为学生群体，还应考虑社会期许偏差：可采用匿名化、强调研究不涉及评价、以及使用区间报告来降低低报风险。

For exposure measurement, self-reported playtime can be triangulated with device screen-time dashboards. Implementation need not collect full screenshots: participants can enter numeric totals and upload a redacted cropped region for spot-checking. For multi-game players, record playtime per title and sum, while also capturing event-week intensity and longest continuous session to separate totals from peaks. In student populations, social desirability can induce under-reporting; anonymization, non-judgment framing, and interval-based reporting can reduce this risk.

偏倚控制上，建议在方案阶段就把关键混杂列为“必须测量”而非“尽量测量”。在本主题中，饮水、纤维来源、睡眠、步数与压力属于一级混杂；咖啡因、酒精、辛辣摄入、泻药使用与月经周期（若相关）可视为二级混杂。若无法全面测量，应在讨论中明确可能残余混杂的方向，并用敏感性分析（例如 E-value 或负对照结局）提示结论的稳健程度。

For bias control, key confounders should be specified as ‘must measure’ rather than ‘nice to have.’ Here, water, fiber sources, sleep, steps, and stress are primary confounders; caffeine, alcohol, spicy intake, laxative use, and menstrual cycle timing (when relevant) are secondary. If full measurement is infeasible, the discussion should explicitly describe residual confounding directions and use sensitivity tools (e.g., E-values or negative-control outcomes) to indicate robustness.

S3. 扩展分析：模型选择、非线性、交互与误差敏感性（Extended analysis notes）

若获得逐日记录数据，分析应优先采用混合效应模型以体现个体内结构：对连续结局（如如厕时长、BSFS）可用线性混合模型；对计数结局（每日排便次数）可用负二项或 Poisson 混合模型；对二元结局（硬便）可用混合效应 Logistic。这样的模型能把“人和人之间的差异”与“同一个人随时间的波动”分开，从而更接近‘当一个人玩得更多时，他/她的排便是否随之改变’这一问题。

With daily repeated measures, mixed-effects models should be prioritized. Continuous outcomes (toilet duration, BSFS) can use linear mixed models; counts (daily bowel movements) can use Poisson or negative-binomial mixed models; binary outcomes (hard stool) can use mixed-effects logistic regression. This separates between-person differences from within-person fluctuation and aligns with the counterfactual question: when an individual plays more, does their bowel behavior change accordingly?

非线性检验建议在方案中预先指定。限制性立方样条可以检验是否存在阈值（例如活动周冲刺后出现突增），也能避免把真实的曲线关系强行拟合为直线。若样条提示阈值，可以进一步用分段回归或按“活动周/非活动周”拆分分析，报告两段的效应差异。在图形呈现上，优先报告预测概率或预测均值曲线，而不是仅给出 p 值。

Nonlinearity checks should be pre-specified. Restricted cubic splines can test thresholds and avoid forcing curved relationships into a straight line. If a threshold is suggested, follow up with piecewise regression or stratification by event vs non-event weeks and report segment-specific effects. For visualization, prioritize predicted means/probabilities rather than p-values alone.

交互项的优先级可按理论路径排序：乙游时长×短睡眠、乙游时长×低步数、乙游时长×低饮水、乙游时长×高如厕刷手机评分。若交互显著，建议使用分层预测图展示，而不是在正文堆叠多个交互系数。中介分析方面，如厕刷手机评分可以作为行为中介，但需要遵守时间顺序并避免强因果措辞；更稳妥的表达是“在统计上解释了部分关联”。

Interaction terms can be prioritized by theoretical pathways: playtime×short sleep, playtime×low steps, playtime×low water intake, and playtime×high phone-on-toilet score. If interactions are significant, present stratified prediction plots rather than stacking coefficients. For mediation, phone-on-toilet can be treated as a behavioral mediator, but temporal ordering should be respected and causal language avoided; a safer phrasing is that it statistically explains part of the association.

测量误差敏感性分析常被忽略，但对屏幕时长与饮食摄入尤其重要。可考虑采用“区间报告+置信度评分”或“重复短问卷”来估计误差范围，并在统计上使用回归校正或 SIMEX 等方法（若研究团队具备能力）。至少，应在附录中展示自报时长与系统统计时长的一致性相关系数与偏差分布，帮助读者判断暴露误差的方向。

Measurement-error sensitivity analyses are often neglected but are important for screen time and diet. Interval reporting with confidence ratings or repeated short questionnaires can estimate error magnitude, and regression calibration or SIMEX can be used when feasible. At minimum, append correlations and bias distributions between self-reported and OS-reported screen-time totals so readers can assess the likely direction of exposure error.

S4. 扩展讨论：反事实、因果边界与研究设计升级（Extended discussion notes）

在讨论因果时，建议明确反事实问题：如果同一个人在其他条件不变时减少乙游时长，他/她的排便是否会改善？观察性关联无法直接回答这一点，因此论文应把结论限定为“在控制可测混杂后仍存在稳定关联”。同时，应提醒读者：排便行为高度受个体基线、饮食底物与情绪状态影响，短期波动并不等同于疾病。

When discussing causality, it helps to state the counterfactual: if the same individual reduced otome playtime while holding other factors constant, would bowel outcomes improve? Observational associations cannot directly answer this, so conclusions should be limited to stable associations after adjusting for measured confounders. Readers should also be reminded that bowel behavior is shaped by baseline traits, dietary substrates, and affect, and short-term fluctuations are not equivalent to disease.

更接近因果的设计可以分三类：自然实验、随机化微干预与个体内交叉。自然实验可利用活动开启或版本更新作为外生冲击，比较更新前后短期排便指标变化；随机化微干预可以分配“如厕不带手机”或“活动周睡眠延时提示”，检验如厕时长是否缩短；个体内交叉则通过长周期跟踪，把同一参与者的高时长周与低时长周进行对照。这些设计的共同点是尽量减少稳定个体差异带来的混杂。

More causal designs fall into three types: natural experiments, randomized micro-interventions, and within-person crossovers. Natural experiments can use event launches or version updates as exogenous shocks to compare short-term changes. Randomized micro-interventions can assign ‘no-phone-on-toilet’ or ‘sleep-extension prompts during event weeks’ and test whether toilet duration decreases. Within-person crossovers compare an individual’s high-playtime weeks against their low-playtime weeks over a longer horizon. The shared goal is reducing confounding from stable individual differences.

伦理表达同样重要。由于主题可能被误读为对某一兴趣群体的道德评判，论文应强调研究目的在于理解行为结构与健康指标之间的统计关系，而非对玩家做价值判断。建议在讨论中加入“去污名化”的措辞：任何高沉浸数字娱乐都可能与久坐和作息扰动共存，关键在于可操作的健康管理，而不是标签化某类游戏。

Ethical framing matters because the topic can be misread as moral judgment of a fandom. The manuscript should emphasize that the goal is understanding statistical links between behavioral structure and health indicators, not evaluating players. Adding explicit de-stigmatizing language helps: many high-immersion digital activities can co-occur with sedentary routines and disrupted sleep; the focus should be on actionable health management rather than labeling a specific genre.

S5. 扩展附录：核心量表条目、编码规则与清洗示例（Extended appendix items）

为便于读者直接复用，本节给出可落地的量表与编码示例。（1）乙游时长：分别填写过去 7 天、过去 30 天、以及最近一次活动周期间的总游玩时长（小时）；若多款乙游，分别填报并求和。（2）活动周强度：过去 14 天中处于“活动冲刺/刷资源”状态的天数（0–14）与主观强度（1–5）。（3）如厕刷手机：过去 14 天中如厕时使用手机的频次（0–100%）、每次平均使用时长（分钟）；换算为 0–10 分。（4）排便困难：用力程度（0–3）、排不尽感（0/1）、便意延迟（有便意但延后>10 分钟，0/1）。（5）梗语沉浸指数：对“上头、我推、吃谷、保底、爆肝、回血、梦女、抽卡”在过去 7 天的使用频率评分（0=从不，1=偶尔，2=经常，3=几乎每天），求和得到 0–24 分。

To facilitate reuse, this section provides implementable items and coding examples. (1) Otome playtime: report total hours for past 7 days, past 30 days, and the most recent event week; sum across titles if multiple games are played. (2) Event-week intensity: number of days (0–14) spent ‘grinding/events’ plus subjective intensity (1–5). (3) Phone-on-toilet: frequency of phone use during toileting (0–100%) and mean minutes per episode, mapped to a 0–10 score. (4) Evacuation difficulty: straining (0–

3), incomplete evacuation (0/1), and urge delay (urge but delayed >10 minutes, 0/1). (5) Meme immersion index: rate usage frequency of eight lexicon items over the past 7 days (0–3) and sum to 0–24.

编码示例：若某参与者“上头=3、我推=2、吃谷=1、其余=0”，则梗语沉浸指数=6。若其如厕手机使用频次为 80%、平均每次 8 分钟，可把频次映射为 8 分、时长映射为 6 分，再取平均或加权得到总分（例如 7.2）。研究者应在方案中明确映射规则，避免事后调整。对暴露变量，建议同时保留连续形式（每+10 小时/周）与分位组形式（四分位），用于主分析与易读的表格呈现。

Coding example: if a participant reports shangtou=3, wo-tui=2, chi-gu=1, others=0, the meme immersion index equals 6. If phone-on-toilet frequency is 80% with 8 minutes per episode, map frequency to 8/10 and duration to 6/10, then average or weight to obtain a total score (e.g., 7.2). Mapping rules should be specified in the protocol. Keep both continuous scaling (per +10 h/week) and quantile groups (quartiles) for modeling and readable tables.

清洗规则示例（需预先定义）：(a) 若 BM/week=0 但填写了 BSFS 或用力评分，标记为不一致并要求核对；(b) 若单次如厕时长>60 分钟，标记为极端值，进入敏感性分析；(c) 饮水量<200 mL/日或>6000 mL/日标记为疑似误填；(d) 纤维摄入为 0 且连续多日硬便者，提示检查饮食记录；(e) 化学指标若低于 LOD 或出现批次异常（如超过批次中位数±5 IQR），按预设规则处理。附录应报告剔除数量与原因，避免“看见结果后才删数据”。

Example cleaning rules (to predefine): (a) if BM/week=0 but BSFS or straining is recorded, flag inconsistency; (b) if toilet duration >60 minutes, flag as extreme and include in sensitivity analyses; (c) water intake <200 mL/day or >6000 mL/day suggests mis-entry; (d) fiber recorded as zero alongside multiple days of hard stool should trigger record checks; (e) chemistry values below LOD or batch anomalies (e.g., beyond median ±5 IQR) should be handled per protocol. Appendices should report exclusions and reasons rather than ‘deleting after seeing results.’

S6. 报告清单与写作结构建议（Reporting checklist and writing structure）

为使稿件更接近可投稿文本，建议用观察性研究的报告清单组织细节：在方法中明确纳入排除、缺失值处理、混杂控制策略与敏感性分析；在结果中提供样本流程图、主要结局的原始分布与调整后估计；在讨论中分层写清机制解释、替代解释与可推广性边界。写作上可把每一段的中心句限定为“我们观察到什么—可能原因是什么—证据强度与限制是什么”，避免段落堆叠概念而缺乏可检验的落点。

To move closer to a submission-ready text, organize details using reporting checklists for observational studies: specify inclusion/exclusion, missing-data handling, confounder control, and sensitivity analyses in Methods; provide a flow diagram, raw distributions, and adjusted estimates in Results; and clearly separate mechanisms, alternative explanations, and generalizability limits in Discussion. A practical writing rule is to structure each paragraph around what was observed, what may explain it, and what the evidence strength and limitations are.

对于双语呈现，建议在最终投稿前根据目标期刊要求选择“主文单语+补充双语”或“段落并列双语”。若保留段落并列双语，应确保两种语言的段落边界对应一致，避免一段中文对应多段英文导致读者误判遗漏。模板化写作的优势在于结构清晰，但也要避免过度重复；可通过在英文部分更强调逻辑连接、在中文部分更强调语境解释来体现互补。

For bilingual presentation, choose either ‘monolingual main text with bilingual supplements’ or ‘parallel bilingual paragraphs’ based on journal requirements. If keeping parallel paragraphs, ensure boundary alignment so readers do not assume missing content. Templates help maintain structure, but avoid excessive repetition; one approach is emphasizing logical connectors in English while using Chinese to provide contextual explanation, making the two complementary.

S7. 扩展生理背景：便形、含水率与代谢指标的写作解释框架（Physiology framing）

在不把化学指标当作诊断的前提下，论文仍可以用更规范的方式解释其“为什么值得测”。第一步是把最直观的表型放在前面：含水率与便形、如厕时长之间存在直觉上的一致性——粪便越干，往往越难在短时间内排出，也更可能表现为 BSFS 偏低与坐厕时间延长。第二步是说明 pH 的角色：粪便 pH 并不是单一因果驱动，而是发酵底物、酸性产物（SCFAs）与含氮产物共同作用的结果；当可发酵碳水不足、蛋白发酵比例上升或肠道停留时间改变时，pH 可能出现轻度上移。第三步才讨论 SCFAs 与氨：SCFAs（尤其丁酸）常被用作“纤维发酵与结肠能量供给”的间接指标，而氨可被视为蛋白发酵或含氮代谢增加的信号。写作时更稳妥的表达是“与某类代谢模式相容”，而不是“证明发生了某一机制”。

Even when chemistry markers are not used diagnostically, the manuscript can explain why they are worth measuring in a more standard way. Step one is to foreground the most intuitive phenotype: fecal water aligns with stool form and toilet duration—drier stool is often harder to pass quickly, corresponding to lower BSFS types and longer sitting time. Step two is clarifying the role of pH: fecal pH is not driven by a single cause but reflects the joint influence of fermentable substrates, acidic products (SCFAs), and nitrogenous products; when fermentable carbohydrates are limited, protein fermentation rises, or transit time changes, pH may shift upward modestly. Only then should SCFAs and ammonia be discussed: SCFAs (especially butyrate) are commonly framed as indirect indicators of fiber fermentation and colonic energy supply, while ammonia can signal increased nitrogen metabolism. A safer writing stance is ‘compatible with a metabolic pattern’ rather than ‘proving a mechanism.’

需要避免的误区是把丁酸或氨的单点变化解读为“好/坏”的线性结论。真实数据中，这些指标受饮食底物、个体菌群结构、采样时间与保存条件影响很大，且不同指标之间可能出现不一致（例如丁酸不变但 pH 上移）。因此更合适的写法是：先描述观测到的方向组合，再给出两个以上的可能解释路径（底物变化、停留时间变化、保存偏差），并明确指出哪一种解释需要额外数据才能区分（例如更细的纤维来源、一次菌群测序或更严格的冷链信息）。

A common pitfall is treating single-marker changes (butyrate or ammonia) as linear ‘good/bad’ conclusions. In real data, these markers are strongly influenced by substrates, baseline microbiome structure, sampling timing, and storage, and markers may disagree (e.g., stable butyrate but higher pH). A more appropriate approach is to describe the observed directional bundle, propose multiple plausible pathways (substrate shifts, transit-time changes, storage artifacts), and explicitly state what additional data would be needed to distinguish them (e.g., detailed fiber sources, a microbiome assay, or stricter cold-chain metadata).

当论文希望把化学指标与行为表型连起来时，可以采用“证据梯度”的叙述：行为层证据最强（因为样本量大、记录频繁），化学层证据较弱（因为子样本小、批次影响大），因此化学结果更适合作为“对机制叙述的约束条件”。例如，若硬便比例上升却观察不到含水率下降趋势，就应在讨论中强调该表型可能更多来自动力不足或如厕延迟，而非水分变化；反之，若含水率与便形方向一致，则可以更自然地讨论饮水/纤维与生活方式的共同作用。

When linking chemistry to behavior, a ‘gradient of evidence’ narrative helps: behavioral evidence is strongest (larger n, frequent records), while chemistry evidence is weaker (smaller subsample, batch sensitivity), so chemistry functions as a constraint on mechanistic storytelling. For

example, if hard-stool prevalence increases but fecal water does not trend downward, the discussion should emphasize that the phenotype may reflect motility or urge-delay factors rather than dehydration. Conversely, consistent directions between water content and stool form support a more natural discussion of the joint roles of hydration, fiber, and lifestyle.

S8. 扩展结果写作：从表格到可读的文字叙述（Results writing patterns）

不少稿件“有表没话”，读者需要在表格里自行拼结论。更稳妥的写法是把每张表对应一个明确的句群：第一句给出总体趋势（随四分位递增/递减）；第二句给出关键对比的量级（Q4 vs Q1 的差异或每 +10 小时的回归系数）；第三句解释调整后是否仍稳定、以及哪个协变量最能削弱效应（例如加入如厕刷手机评分后系数衰减）。对于二元结局，建议同时报告预测概率或绝对风险差，以避免 OR 被误读为“风险倍增”。

Many papers ‘have tables but no story,’ forcing readers to infer conclusions. A safer pattern is to map each table to a tight sentence set: sentence 1 states the overall trend (increasing/decreasing across quartiles); sentence 2 quantifies the key contrast (Q4 vs Q1 difference or the per +10h regression coefficient); sentence 3 notes whether the adjusted association remains stable and which covariate most attenuates it (e.g., adding phone-on-toilet reduces the coefficient). For binary outcomes, report predicted probabilities or absolute risk differences alongside odds ratios to avoid misreading ORs as direct risk multipliers.

如果希望突出“活动周”这一特征，可以在结果中加入一个简洁的交互或分层段落：先说明活动周比例或强度随时长上升；再报告活动周与非活动周的估计差异；最后给出一句解释（活动周同时压缩睡眠与增加久坐，因此可能放大效应）。注意，交互结果不宜堆砌多个 p 值，最好用一张图或一组预测概率把信息浓缩。

If ‘event weeks’ are central, include a short interaction or stratified paragraph: first show that event-week intensity rises with playtime; then report estimate differences between event and non-event weeks; finally provide a one-sentence interpretation (event weeks co-compress sleep and increase sedentary time, potentially amplifying effects). Avoid stacking p-values; a single plot or predicted-probability panel usually communicates interactions better.

对于“梗语沉浸指数”，结果段落的写法可更偏描述性：强调其随时长呈梯度上升，提示其可作为参与结构的刻画；若把其纳入模型，可报告其是否改变乙游时长的主效应（作为混杂/中介/修饰的可能）。写作上应避免把亚文化词汇当作笑点，而应把它当作一种可量化的社会行为线索，语气保持中性。

For the meme immersion index, results can be more descriptive: emphasize its graded increase with playtime, suggesting it captures engagement structure; if included in models, report whether it alters the main playtime effect (as a potential confounder, mediator, or modifier). Avoid treating subcultural terms as jokes; present them neutrally as quantifiable social-behavioral signals.

S9. 扩展局限性：偏倚来源与如何在论文中诚实表达（Limitations in practice）

局限性写作的关键不是罗列“可能有偏倚”，而是具体说明偏倚会把结果推向哪个方向。信息偏差方面，若高时长玩家更倾向低报时长（社会期许），则真实效应可能被低估；若高时长玩家更敏感、更愿意记录排便不适，则效应可能被高估。选择偏差方面，愿意参与粪便采样者可能更健康或更关注肠道问题，化学子样本的结论可能无法代表全部玩家。混杂方面，压力与睡眠往往同时影响游戏与排便，若测量粗糙，残余混杂可能仍然存在。把这些方向写清楚，比“可能有偏倚”更能让审稿人相信作者理解自己的数据。

A good limitations section does more than list ‘possible bias’; it states which direction a bias would push estimates. For information bias, if high-playtime participants under-report due to social desirability, effects may be underestimated; if they are more symptom-aware and more likely to record discomfort, effects may be overestimated. For selection bias, stool-sample providers may be healthier or more gut-focused, limiting representativeness of chemistry findings. For confounding, stress and sleep influence both gaming and bowel outcomes; coarse measurement leaves residual confounding. Stating directionality demonstrates that authors understand their data beyond generic caveats.

批次效应与实验室误差是化学指标的特有局限。若不同批次对应不同时间招募、不同地区或不同活动周强度，则批次可能与暴露相关，导致伪关联。论文中应明确是否做了批次调整、是否随机化检测顺序、以及是否有重复样本/空白对照。即便无法完全消除，也应把其作为解释边界的一部分：化学结论属于探索性，需要更严格的实验设计复核。

Batch effects and lab error are specific limitations for chemistry markers. If assay batches correlate with recruitment time, region, or event-week intensity, batches can correlate with exposure and induce spurious associations. The manuscript should state whether batch adjustment was used, whether assay order was randomized, and whether replicates/blanks were included. Even if not fully eliminated, position chemistry findings as exploratory and in need of more controlled follow-up.

最后，语言边界也属于“方法学局限”：研究对象来自中文互联网文化场域，梗语指数与参与结构的测量可能难以直接迁移到其他语言社群。论文可将此表述为“构念等价性”问题，并建议未来在不同文化语境中重新开发等价量表，而不是简单翻译词汇。这样的表述能把亚文化变量从‘趣味点’提升为严肃的测量学议题。

Finally, language and culture are methodological limits: the lexicon-based engagement measures arise from Chinese internet contexts and may not transfer directly to other language communities. Frame this as a construct-equivalence issue and recommend redeveloping equivalent scales in other cultural settings rather than literal translation. This elevates subcultural variables from ‘fun context’ to a serious measurement problem.

S10. 扩展伦理与沟通：如何避免污名化并提供对读者友好的结尾（Ethics and communication）

由于研究主题涉及特定兴趣群体，论文的沟通策略应尽量避免“把责任推给兴趣”。更合适的公共健康叙事是：任何高沉浸数字活动（社交媒体、短视频、追剧、电竞）都可能与久坐、饮水不足与睡眠延迟共存；乙游只是其中一种具有代表性的结构。把乙游作为案例的价值在于其时间结构清晰、活动周波动明显、且社区参与可被量化，从而有利于方法学演示。

Because the topic touches a specific interest community, communication should avoid ‘blaming the hobby.’ A more appropriate public-health narrative is that many high-immersion digital activities (social media, short videos, binge watching, esports) can co-occur with sedentary time, low hydration, and delayed sleep; otome games are one representative structure. Their value as a case lies in clear time structure, pronounced event-week volatility, and quantifiable community participation, which support methodological demonstration.

对读者友好的结尾可以采用“三步建议”格式：第一步是提醒常识性健康行为（规律作息、增加步行、足量饮水与膳食纤维）；第二步是给出最小干预单元（如厕不刷手机、限制坐厕时间）；第三步是给出警示信号与就医边界（持续便秘、便血、明显腹痛、体重下

降）。这种结尾既能体现研究的公共价值，也能避免把统计关联误读为医疗建议。

A reader-friendly ending can use a three-step guidance format: (1) reinforce general health behaviors (regular sleep, more walking, adequate water and fiber); (2) propose a minimal intervention unit (no-phone-on-toilet, limit sitting time); and (3) state red-flag symptoms and clinical boundaries (persistent constipation, blood in stool, severe pain, weight loss). This communicates public value while reducing the risk that statistical associations are mistaken for medical advice.

S11. 扩展数据结构：从原始日记到分析数据集（Data structure notes）

日记原始数据通常以“长表（long format）”存储：每行对应某一天或某一次排便事件。为了进入回归模型，需要做两步整理：第一步是把事件表聚合为个体-天层面的派生变量（当日排便次数、当日最大/平均如厕时长、当日硬便指示、当日如厕手机总时长）；第二步是把 14 天聚合为个体层面的平均暴露与平均结局（例如 BM/week、平均如厕时长、硬便比例），并保留波动指标（标准差、最大值、活动周标记）。在论文中给出一个简短的数据流示意（原始字段→派生字段→分析字段）会显著提升可复现性。

Diary raw data are typically stored in long format, where each row corresponds to a day or a bowel-movement event. Moving into regression requires two transformations: (1) aggregate the event table into person-day derived variables (daily BM count, daily max/mean toilet duration, daily hard-stool indicator, daily total phone-on-toilet time); (2) aggregate 14 days into person-level mean exposures and outcomes (BM/week, mean duration, hard-stool prevalence) while preserving variability metrics (SD, maxima, event-week flags). Including a short data-flow schematic (raw fields → derived fields → analysis fields) materially improves reproducibility.

对“复合便秘样”指标，建议在附录明确其构成的逻辑表达式，并说明其目的在于提升表型稳定性而非替代临床诊断。例如：Composite=1 若（BM/week<3）或（硬便比例≥某阈值）或（平均如厕时长≥12 分钟）或（用力评分≥2 且排不尽=1）。不同研究可根据目标人群调整阈值，但必须在分析前固定，并在敏感性分析中检验阈值变化是否改变方向。

For the composite constipation-like indicator, explicitly define the logical expression in an appendix and state its purpose as phenotype stabilization rather than clinical diagnosis. For example: Composite=1 if BM/week<3 OR hard-stool prevalence above a threshold OR mean toilet duration ≥12 minutes OR (straining ≥2 AND incomplete evacuation=1). Thresholds can be tuned by population, but they should be fixed before analysis and tested via sensitivity analyses to confirm directional stability.

对于缺失值，日记数据常见两种机制：随机漏填与系统性漏填（例如活动周更忙导致漏填）。若缺失与暴露相关，完全案例分析可能产生偏倚。实践上，可采用多重插补处理日汇总层变量（饮水、睡眠、压力），对事件层变量可采用规则化补全（例如缺失时长但有排便事件，可用该个体的中位数时长替代并做敏感性分析）。无论采用何种方法，都应在正文中报告缺失比例，并在附录给出插补模型的核心变量。

Missingness in diaries commonly arises as random omissions or systematic omissions (e.g., busier event weeks). If missingness correlates with exposure, complete-case analysis can bias estimates. Practically, multiple imputation can be used for daily-summary variables (water, sleep, stress), while event-level variables can use rule-based filling (e.g., missing duration but a BM event exists—impute using that individual's median and run sensitivity analyses). Regardless of method, report missingness rates in the main text and describe key imputation variables in an appendix.

S12. 扩展审稿视角：常见质疑与可预先准备的回应（Reviewer-facing notes）

围绕该主题的常见质疑通常集中在三点：第一，“游戏类型是否只是噪声？”对此可以回应：研究关注的是时间结构与生活方式耦合，乙游作为具有活动周脉冲与强叙事陪伴的代表性产品，有助于构建可检验的暴露框架；同时，模型并不假设乙游具有独特生物效应。第二，“排便时长是否只是刷手机？”对此可以回应：我们显式测量并调整如厕手机使用，并建议未来加入用力与排不尽等更接近生理困难的终点。第三，“化学指标是否可靠？”对此可以回应：化学结果被定位为探索性一致性证据，并提供批次调整与质控信息。

Common reviewer concerns cluster around three points. (1) 'Is game genre just noise?' Respond that the focus is time structure and lifestyle coupling; otome games, with event-week pulses and narrative companionship, provide a testable exposure framework without assuming unique biological effects. (2) 'Is toilet duration just phone scrolling?' Respond that phone-on-toilet behavior is explicitly measured and adjusted, and future work can add straining and incomplete evacuation to better capture physiological difficulty. (3) 'Are chemistry markers reliable?' Respond that chemistry is positioned as exploratory consistency evidence and is accompanied by QC and batch-adjustment information.

此外，审稿人可能要求更强的因果表述。较稳妥的准备是提前写好“因果边界段落”：说明观察性研究无法排除反向因果与残余混杂，因而结论限定为关联；并提出下一步设计路线（自然实验、随机化微干预、个体内交叉）。如果作者能把“下一步怎么做”写得具体，往往比在当前数据上硬推因果更能说服审稿人。

Reviewers may also push for stronger causal claims. A safer preparation is a 'causal-boundary paragraph': state that observational designs cannot rule out reverse causation or residual confounding, so conclusions are associative, and outline concrete next-step designs (natural experiments, randomized micro-interventions, within-person crossovers). Clear 'what to do next' plans are often more persuasive than forcing causality from current data.

S13. 扩展暴露刻画：从“总时长”到“节律与峰值”（Exposure decomposition）

仅用“小时/周”刻画乙游暴露会把不同节律的玩家混在一起。更细的刻画至少应包含四个维度：总量（周总时长）、节律（每天登录次数与分布，例如睡前集中或全天分散）、峰值（单次连续使用最长时长、活动周增量时长）、以及场景（如厕/床上/通勤等主要使用地点或时段）。在实际问卷中，可以用少量条目近似：过去 7 天平均每天打开乙游的次数、最近一次活动周每天额外增加的时长、以及最常见的三种使用场景。把这些维度与排便结局并列呈现，有助于读者理解“时长背后的生活方式结构”。

Using hours per week alone mixes players with different rhythms. A richer exposure characterization should include at least four dimensions: total volume (weekly hours), rhythm (number and distribution of logins per day, e.g., bedtime-concentrated vs dispersed), peaks (longest continuous session, event-week incremental hours), and context (primary usage settings such as toileting, bed, or commuting). In practice, a short questionnaire can approximate these dimensions: mean daily opens over the past 7 days, extra time per day during the most recent event week, and the three most common contexts. Presenting these dimensions alongside bowel outcomes helps readers interpret the lifestyle structure behind 'hours.'

在统计上，节律与峰值变量可以帮助解释为何同样的总时长可能产生不同的健康关联：例如，睡前集中使用更可能压缩睡眠，活动周峰值更可能叠加久坐与饮食紊乱，而分散的短时登录可能更多反映碎片化刷屏（包括如厕刷手机）。因此，未来实证研究可以采用两步模型：第一步用总时长预测结局；第二步加入节律/峰值/场景变量观察主效应是否衰减，从而判断"总时长"究竟是代理变量还是具有独立信息。

Statistically, rhythm and peak variables help explain why identical total hours can link to different health associations: bedtime-concentrated use is more likely to compress sleep; event-week peaks more likely co-occur with sedentary time and dietary disruption; and dispersed micro-sessions may reflect ubiquitous scrolling (including phone use on the toilet). Future empirical work can use a two-step model: first regress outcomes on total hours; then add rhythm/peak/context variables to see whether the main effect attenuates, indicating whether total hours are a proxy or carry independent information.

如果研究团队能够获得更客观的行为日志（例如系统屏幕使用、应用前后台切换次数、或游戏内活动记录），还可以把"暴露刻画"升级为时间序列特征工程：计算活动周前后的突增幅度、夜间使用比例、如厕时段使用比例、以及连续使用的分布。此类特征既能减少回忆偏差，也更适合与混合效应模型、因果推断框架（如断点回归或差分中的差分）结合。但在隐私与参与者负担的约束下，问卷+日记的轻量刻画仍具有现实可行性。

If objective logs are available (OS screen time, app foreground switches, or in-game activity logs), exposure characterization can be upgraded into time-series features: event-week surge magnitude, nighttime-use proportion, toileting-time proportion, and distributions of continuous sessions. These features reduce recall bias and pair naturally with mixed models and quasi-causal designs (regression discontinuity, difference-in-differences). Under privacy and burden constraints, however, questionnaire-plus-diary remains a feasible lightweight approach.

S14. 扩展临床边界：研究指标与就医信号的区分（Clinical boundaries）

由于排便相关话题容易引发读者焦虑，论文在结尾或补充材料中应明确"研究指标≠临床诊断"。在年轻成人中，排便频次与便形存在正常波动，短期内的硬便或排便次数下降不必然意味着疾病。研究的公共健康价值主要在于提示生活方式结构与功能性表型之间可能存在可调节的关联，而不是提供个人医疗建议。

Because bowel topics can trigger anxiety, the manuscript should clearly state that research indicators are not clinical diagnoses. Among young adults, bowel frequency and stool form vary within normal ranges, and short-term hard stool or fewer bowel movements do not necessarily indicate disease. The public-health value lies in suggesting potentially modifiable links between lifestyle structure and functional phenotypes, not in delivering individualized medical guidance.

同时也应列出需要就医评估的"红旗信号"，以防读者误把模板当成健康指南：例如持续便秘且伴随便血或黑便、明显体重下降、持续发热、夜间痛醒、严重腹痛、贫血提示、或家族史相关的风险增加等。写入这些边界并不会削弱论文，相反会强化其专业性与责任感。

It is also appropriate to list red-flag symptoms that warrant medical evaluation so readers do not treat a template as health advice: persistent constipation with blood in stool or black stools, marked weight loss, persistent fever, nocturnal pain, severe abdominal pain, signs of anemia, or relevant family history. Including boundaries strengthens professionalism and responsibility rather than weakening the manuscript.

最后，对干预建议也应保持"低风险、可普适"的原则：规律作息、增加活动、足量饮水、提高膳食纤维、减少如厕刷手机与久坐停留，这些建议对多数人群是安全且可能有益的；但若存在基础疾病或正在使用影响肠道的药物，个体应以医生建议为准。研究文本中的建议应以这种方式呈现，避免越界。

Finally, any behavioral suggestions should follow a low-risk, broadly applicable principle: regular sleep, more activity, adequate water, higher dietary fiber, and limiting phone-on-toilet sitting are generally safe and potentially beneficial for many people. However, individuals with underlying conditions or on gut-affecting medications should follow clinician advice. Recommendations should be framed in this boundary-aware way to avoid overreach.

从写作规范角度，建议在摘要与结论中始终使用"关联/相关（associated with）"等中性词，并把"可能机制"用条件句表达（may, might），同时在首次出现便秘相关定义时注明其为研究操作性定义。这样既符合观察性研究的语言习惯，也能降低读者把统计结果当作因果宣告的风险。若期刊允许，可增加一段简短的通俗摘要（plain-language summary），用更少术语解释研究发现与建议。

From a reporting-standards perspective, use neutral language such as 'associated with' in abstracts and conclusions, phrase mechanisms conditionally (may/might), and clarify that constipation-related definitions are operational for research. This aligns with observational-study norms and reduces the risk that readers interpret statistics as causal declarations. If permitted, add a short plain-language summary to explain findings and suggestions with fewer technical terms.

若研究面向青少年或高压学生群体，还应在讨论中提示"时间管理"与"自我照护"之间的平衡：与其要求完全戒断，不如倡导在活动周设定可达成的边界（例如固定睡眠底线、每小时起身、如厕不刷手机），并把这些边界视为支持学习与健康的工具，而不是道德约束。这样的表述更容易被学校与家庭接受，也更符合公共健康干预的现实路径。

If the study targets adolescents or high-pressure students, the discussion can also note the balance between time management and self-care. Instead of advocating total abstinence, promote achievable boundaries during event weeks (a sleep floor, hourly standing breaks, no-phone-on-toilet) framed as tools supporting learning and health rather than moral constraints. This framing is often more acceptable to schools and families and aligns with practical public-health intervention pathways.

总之，本模板的写作目标是：在保持科学语气与方法透明的前提下，把一个看似"生活化"的暴露转化为可测量、可建模、可解释的研究问题，并为后续真实数据研究提供直接可替换的结构。

In sum, this template aims to translate an apparently 'everyday' exposure into a measurable, modelable, and interpretable research question while maintaining scientific tone and methodological transparency, and to provide a structure that can be directly substituted with real-world data in future studies.

在最终排版阶段，建议对照目标期刊的格式要求检查：摘要结构、关键词数量、表图标题格式、参考文献格式与单位书写一致性，并确保双语段落的顺序与对应关系清晰。

At final typesetting, check the target journal's requirements for abstract structure, keyword count, table/figure caption format, reference style, and unit consistency, and ensure bilingual paragraph order and correspondence are clear.

如需进一步扩展，可把各量表条目的中英文版本以表格形式列出，并补充示例截图与填报说明。

If further expansion is needed, list bilingual instrument items in a table and add example screenshots and completion instructions.

以上补充段落可按投稿要求保留或删减。

These supplementary paragraphs can be kept or trimmed based on submission requirements.